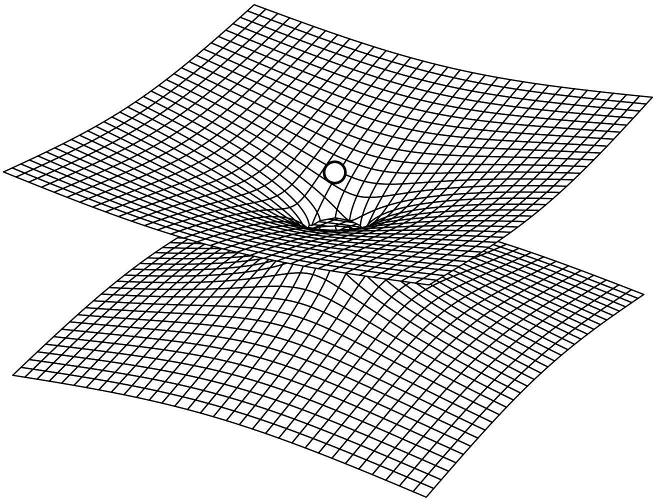

Lecture Notes on General Relativity

Physics 225

Vercil Juan

IV - BS Physics

1st Semester, AY 2026-2027

$$\Gamma^{\rho}_{\mu\nu} = \frac{1}{2} g^{\rho\sigma} (\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu})$$

$$\frac{d^2 x^\sigma}{d\lambda^2} + \Gamma^{\sigma}_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} = 0$$

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

"Rationem vero harum Gravitatis proprietatum ex Phenomenis nondum potui deducere, et Hypotheses non fingo."

– I. Newton

Physics 225 - General Relativity 1
Course Information

Course Description

Description: Manifolds, modern differential geometry and tensor analysis; basic principles of general relativity; Einstein field equations and their mathematical properties; exact solutions; linearized theory; variational principles and conservation laws; equations of motion; gravitational waves; experimental tests.

Instructor Details

Instructor: Dr. Michael Francis Ian Vega

Email: ivega.upd.edu.ph

Lecture Hours: WF, 10:00 - 11:30 AM

Course Requirements

Problem Sets Problem sets will be assigned throughout the semester, ideally one per week. These will be submitted electronically using a Google Form that I will be sending out close to the deadline. From each problem set a random selection of items will be graded. It is your responsibility to ensure that your work is neat and legible. If we cannot understand your work then you will not get any credit. These problem sets will make up the 30% of your final grade.

Written Exam There will be two written examns in the course. Each exam will constitute 30% of the final grade. These are in-class, open-notes exams. You will have to designate one notebook as your 225 notebook at the start of the semester. Only this notebook may be used to help you during the exams.

Oral Exam This will be scheduled during or shortly after Finals Week. This will make up the remaining 10% of your final grade.

Main References

We shall not have a main textbook for the class. I used *Schutz*, supplemented by the first edition of *d'Inverno*, when I first started learning GR. I think they are both very good texts. However, I have my own way of presenting GR that will not always follow the structure or presentation of any single book. My approach is closest in style to the treatments of *Liang+Zhou* and *Wald*.

There are numerous sources you can refer to. I highly recommend that you look at different texts to help understand my lectures better. Here are some good books I've used on numerous occasions. - I. Vega

Course Syllabus

The Syllabus for the course can be found here:

- Course Syllabus

For General Relativity

- R d'Inverno, J Vickers, **Introducing Einstein's Relativity: A Deeper Understanding**, 2022
- C Liang and B Zhou, **Differential Geometry and General Relativity**, 2023
- R Wald, **General Relativity**, 1984
- V Ferrari, L Gualtieri, P Pani, General Relativity and its Applications, 2021
- T Padmanabhan, Gravitation: Foundations and Frontiers, 2010
- A Zee, Einstein Gravity in a Nutshell, 2013
- S Carrol, **Spacetime and Geometry: An Introduction to General Relativity**, 2003
- Choquet-Bruhat, Introduction to General Relativity, Black Holes, and Cosmology, 2015

- J Hartle, **Gravity: An Introduction to Einstein's General Relativity**, 2003. (undergrad)
- B Schutz, **A First Course in General Relativity 2nd Ed.**, 2009. (undergrad)
- MP Hobson, GP Efstathiou, and AN Lasenby, General Relativity: An Introduction for Physicists, 2006. (strong for cosmology)
- S Hollands, K Sanders, **Lecture Notes on General Relativity**, 2015
- S Weinberg, **Gravitation and Cosmology**, 1972
- A Einstein, **Relativity: The Special and General Theory**, 1920

For Differential Geometry

- B O'Neill, **Semi-Riemannian Geometry**, 1983
- S Lovett, **Differential Geometry of Manifolds 2nd Ed**, 2020
- G Rudolph, M Schmidt, **Differential Geometry and Mathematical Physics: Part I**, 2013

Note :

The hyperlinks in this lecture notes act as follows: The ■ hyperlink, when pressed, opens the relevant `pdf` in my (the Author's) local machine. This only works for my personal computer. The hyperlink on the **name of the book** links to a source of the book on the internet. Press that hyperlink to access that book.

Reminders

Exam Dates

- ☐ Long Exam 1 : **October 12, 2026, 9-12 AM**
- ☐ Long Exam 2 : **December 7, 2026, 9-12 AM**
- ☐ Oral Exam : **Finals week (TBA)**

Problem Set Dates

- ☐
- ☐
- ☐
- ☐
- ☐

Preface

These lecture notes present a systematic development of general relativity based on the material covered in the corresponding course. They are based on the lectures delivered by Dr. Vega and are intended primarily as a record and reference for the concepts, mathematical methods, and physical arguments developed throughout the course. The notes are not verbatim transcripts – although the author has made the utmost effort to preserve the words spoken throughout the lecture; rather, they reconstruct and organize the material presented in class, with additional mathematical details, context, references and explanations where necessary to make the discussion coherent when read independently.

The presentation follows the progression of the course, beginning with the foundations of relativity and the transition from Newtonian gravitation to the geometric description of gravity, and subsequently developing the mathematical framework required for the theory and its physical applications. The notes assume a background in undergraduate physics and mathematics, including classical mechanics, special relativity, multivariable calculus, linear algebra, and differential equations. References and footnotes are provided throughout for readers seeking more detailed, alternative, or advanced treatments of particular topics. This is intended mostly as a personal work for the author's own review, although it has been organized sufficiently to serve as a suppelementary reference such that that it can be shared to peers, friends, or other students when requested.

These lecture notes was prepared by the author using LaTeX 2_{ε}.

August, 2026

Vercil Juan
V - BS Physics
University of the Philippines Diliman

Physica Vincit Omnia

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List of Figures

List of Theorems

Chapter 1

Review of Newtonian Physics

1.1 Introduction

> "That gravity should be innate, inherent, and essential to matter, so that one body may act upon another at a distance through a vacuum, without the mediation of anything else... is to me so great an absurdity that I believe no man who has, in philosophical matters, any competent faculty of thinking can ever fall into it." – NEWTON

We begin this course with the assumption that you have no idea what General Relativity is. How do we quantify what is gravity? What even is gravity? We begin with the classical equations of motion.

The classical equations of motion, first described by Galileo and formalized by Isaac Newton, have long served as the default model for describing the behavior of classical objects under the influence, or lack thereof, of external forces, and remain a fundamental part of physics education due to their success in describing matter within the scope of ordinary human observation. However, in the dawn of the 20th century, contradictions between these classical descriptions and experimental observations began to surface, making it apparent that a new description of reality was needed to reconcile theory with these results.

To motivate ourselves, we describe three concepts introduced over the history of classical mechanics and explain how it leads to certain contradictions or problems that motivated the realization and invention of general relativity

1.1.1 Newton's Law of Gravitation

Newton's Laws as he prescribed them is expressed as follows:

$$m_N \frac{d^2\mathbf{x}_N}{dt^2} = G \sum_M \frac{m_N m_M (\mathbf{x}_M - \mathbf{x}_N)}{\|\mathbf{x}_M - \mathbf{x}_N\|^3} \tag{1.1.1}$$

More often, it is shortened as:

Definition 1.1 : Newton's Law of Gravity

$$\mathbf{F} = -\frac{GMm}{r^2}\hat{\mathbf{r}} \tag{1.1.2}$$

This is the familiar form of Newtonian gravity, and is perhaps one of the most recognizable equations in physics. This is what every high schooler knows and what every mentor presents, and is trivial to derive via experimentation.

We can separate the mass m experiencing the force from the mass M producing it, and write

$$\mathbf{F} = m_g\left(-\frac{GM}{r^2}\hat{\mathbf{r}}\right) \tag{1.1.3}$$

We can henceforth call m_g as the *gravitational mass* and the quantity $-\frac{GM}{r^2}\hat{\mathbf{r}}$ as the *gravitational field*

We can express it more generally by recognizing the force field is simply the negative gradient of a potential field.

$$\mathbf{F} = m_g(-\nabla\Phi) \tag{1.1.4}$$

where $\Phi = -\frac{GM}{r}$ is the gravitational potential produced by the source mass.

Thus, we can henceforth call m_g as the *gravitational charge.* It follows from the fact that

$$\text{charge} \times \text{field gradient} = force \tag{1.1.5}$$

This formulation is universal across all of physics.

Analogous to electromagnetism, where

$$\mathbf{F} = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}\hat{\mathbf{r}} \tag{1.1.6}$$

Like the electric charge q, the gravitational charge m_g determines the strength of the interaction of the forces.

An important property that Newtonian gravitation has is that it has superposition. i.e., fields add together linearly. This is therefore also true of the gravitational potential.

For a system of point masses, the total potential field $\Phi(\mathbf{x})$ at a point $\mathbf{x}$ is given by the sum of the potentials produced by each individual mass:

$$\begin{aligned}\Phi(\mathbf{x}) &= \sum_i \Phi_i \\ &= -G\sum_i \frac{M_i}{|\mathbf{x}-\mathbf{r}_i|}\end{aligned}$$

We can extend this description from a discrete collection of point masses to a continuous mass distribution. Let the mass contained in an infinitesimal volume $d^3\mathbf{r}$ be

$$dm_i = \rho(t,\mathbf{r})\, d^3\mathbf{r} \tag{1.1.7}$$

Then we arrive with the continuum version of the gravitational potential field generated by a mass distribution.

$$\Phi(\mathbf{x}) = -G\int \frac{\rho(t,\mathbf{r})}{|\mathbf{x}-\mathbf{r}|}\, d^3\mathbf{r} \tag{1.1.8}$$

Here, $\mathbf{r}$ denotes the position of each source element of the mass distribution, while $\mathbf{x}$ denotes the point at which the potential is being evaluated.

We can now obtain a differential equation for the gravitational potential. Taking the Laplacian of Φ gives

$$\nabla^2\Phi = -G\int \rho(t,\mathbf{r})\, d^3\mathbf{r}\, \nabla^2\frac{1}{|\mathbf{x}-\mathbf{r}|} \tag{1.1.9}$$

Recall that this operation results to a dirac-delta function

$$\nabla^2 \frac{1}{|\mathbf{x} - \mathbf{r}|} = -4\pi\delta(\mathbf{x} - \mathbf{r}) \tag{1.1.10}$$

which relates the Laplacian of the inverse-distance potential to the Dirac delta function. Substituting this into the previous expression gives

$$\nabla^2 \Phi = 4\pi G \int \rho(t, \mathbf{r})\, d^3\mathbf{r}\, \delta(\mathbf{x} - \mathbf{r}) \tag{1.1.11}$$

Recall the sifting property of the Dirac delta

$$\int f(r)\delta(x - r)\, dr = f(x) \tag{1.1.12}$$

From this we obtain

$$\nabla^2 \Phi = 4\pi G \rho(t, \mathbf{x}) \tag{1.1.13}$$

Since $\mathbf{x}$ is an arbitrary point, we may relabel it as

$$\nabla^2 \Phi(t, \mathbf{r}) = 4\pi G \rho(t, \mathbf{r}) \tag{1.1.14}$$

Thus, we arrived with the fundamental field equation which governs Newtonian Gravity

Theorem 1.1 : Field Equation for Newtonian Gravity

$$\nabla^2 \Phi = 4\pi G \rho \tag{1.1.15}$$

The significance of a field equation is that it relates a field to its sources. In this case, the mass density ρ acts as the source of the gravitational potential field Φ.

Note the following

$$\Phi = -G \int \frac{\rho\, d^3\mathbf{r}}{|\mathbf{x} - \mathbf{r}|} \iff \nabla^2 \Phi = 4\pi G \rho \tag{1.1.16}$$

The two are notionally equivalent. One is a differential equation and the other is the integral form. One can switch to and from them using the Laplacian and Green's functions[1]. It assigns the definition of the field, and fields assign values to every point in space.

[1] Green's functions are essentially the inverse of a laplacian, ∇^{-2}

However, this formulation presents a fundamental problem. Since the potential Φ is determined directly by the mass density ρ, any change in the mass distribution produces an immediate change in the gravitational field throughout space. There is no propagation delay or retardation. In other words, Newtonian gravity permits changes in the gravitational field to propagate instantaneously, which is incompatible with relativistic causality. This problem is one of the motivations for general relativity.

The corresponding requirement in GR is that, in the appropriate weak-field and slow-motion limit, general relativity must reduce to Newtonian gravity.

Now that we have the potential of the field that the source creates, and thus the force, how do we determine how an object moves in response to that field?

1.1.2 Newton's Equations of Motion

The quintessential equation governing the motion of classical bodies was expressed by Newton as

$$\mathbf{F} = m\mathbf{a} \tag{1.1.17}$$

Everybody and their mother knows this shit.

This states that if one sums of all the forces acting on an object, it must equal its mass multiplied by its acceleration:

$$\sum \mathbf{F} = m\mathbf{a} \tag{1.1.18}$$

We can rewrite the equation as

$$\mathbf{a} = \frac{1}{m} \sum \mathbf{F} \tag{1.1.19}$$

Notice that the m term acts as a kind of *resistance* term[2], similar to the concept of inertia. This is to be expected, since Newton's first law can essentially be understood as the statement that, in the absence of a net force, an object does not change its state of motion.

Hence, we name this quantity the *inertial mass*, m_i. Inertia denotes the difficulty or resistance of an object to a change in its state of motion.

[2] Recall Ohm's Law $V = IR$.

We can restate (1.1.19) with m replaced by m_i, to distinguish inertial mass from the gravitational mass introduced previously:

$$\mathbf{a} = \frac{1}{m_i}\mathbf{F} \tag{1.1.20}$$

Recall that we already have the definition of $\mathbf{F}$ from (1.1.4). Thus, to calculate the motion of an object under the influence of gravity, we simply calculate its acceleration:

$$\mathbf{a} = \frac{m_g}{m_i}(-\nabla\Phi) \tag{1.1.21}$$

Notice that this is similar in form to the equation of motion for a charged particle in an electric field.

$$\mathbf{a} = \frac{q}{m_i}\mathbf{E} \tag{1.1.22}$$

where m_i is the inertial mass, the measure of an object's resistance to changes in its motion. In both cases, the coupling to the field determines the force, while the inertial mass determines the resulting acceleration.

However, for calculations involving Newtonian gravity, the ratio of the gravitational charge and inertial mass $\frac{m_g}{m_i}$ is assumed to be 1. Thus, for gravity, the acceleration of a body is expressed as

$$\mathbf{a} = -\nabla\Phi. \tag{1.1.23}$$

This assumes that

$$m_g = m_i, \tag{1.1.24}$$

This feels unnatural and unexpected, a truly remarkable coincidence in Newtonian theory. Why should it be the case that one force acts differently from other forces in terms of its charge carriers? Mass is a fundamental property that affects other forces, there's no immediate evidence why mass is gravity's charge carrier.

There is no *a priori* reason why the magnitude of gravitational force on a particle should equal the quantity that determines the particle's resistance to an applied force in general. They play fundamentally different roles in the theory. It is analogous to assuming that an object's coupling to some other field must be equal to its inertial mass, like an electron's electromagnetic properties being dependent on mass.

Nevertheless, for gravity this equality is accepted experimental fact without explanation.

1.1.3 Weak Equivalence Principle

The assumption that

$$m_i = m_g \tag{1.1.25}$$

is called the *weak equivalence principle*.

It follows that, for objects only affected by gravity, the equation of motion is simply

$$\mathbf{a} = -\nabla\Phi. \tag{1.1.26}$$

That is, any test mass placed in the same gravitational field experiences the same acceleration, regardless of its mass or composition. The nature of the object itself does not appear in the equation of motion. The field prescribes the acceleration, the test body merely responds to it.

One could therefore remove the test object m and nothing about the gravitational field would change. In Newtonian mechanics, this is treated as a universal fact: the gravitational field exists independently of the particular body used to probe it.

However, this raises a deeper question.

One could argue that the stage for the motion of an object is the field itself. If we remove the object from some local region, the field remains.

By the same logic, if we remove the field itself, what remains? In Newtonian mechanics, space $\mathbb{R}^3$ remains as the stage on which the field and matter exist.

The influence of gravity has nothing to do with the nature of the test object itself. The gravitational field does not change when the test object is removed, and its motion depends only on the field at its location.

One might ask: if we remove all mass, would there not be no gravity? The answer is that we must distinguish the sources of the field from the test body. We are considering local interactions. Mass may exist somewhere outside the local region of interest and still generate a gravitational field within it. Given the local potential Φ, we can determine information about its sources, but the test body itself is not one of those sources.

Thus,

$$\mathbf{a} = -\nabla\Phi \tag{1.1.27}$$

is treated as inertial motion in a gravitational field.

In the stage analogy, one could think of $-\nabla\Phi$ as living on the stage of space $\mathbb{R}^3$. The field exists on space and determines how objects move through it. In some fields such as particle physics, that is the case, they treat gravity as a special field which acts on every particle and every particle responds to in the same way

However, in GR, $-\nabla\Phi$ is no longer treated as a field sitting on top of the stage. The gravitational field *is* the stage. The space time is what the rest of physics takes place in.

The strong equivalence principle extends this idea: locally, the laws of physics in a freely falling frame are the same as those of an inertial frame in the absence of gravity. In other words, locally, gravity can be transformed away by choosing an appropriate freely falling frame.

Inertial motion is therefore determined by the structure of spacetime.

Physics takes place on the stage called spacetime, and GR governs that space time. The spacetime structure remains even after the particular body used to probe it is removed.

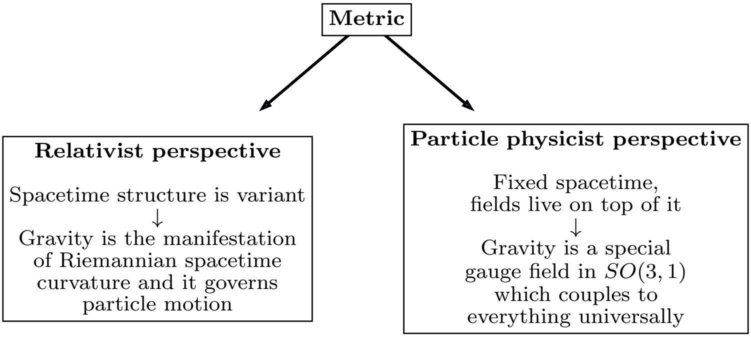
In GR, instead of the gravitational potential field Φ, we have the metric:

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu. \tag{1.1.28}$$

The metric determines the geometry of spacetime and, consequently, the inertial trajectories of objects.

In particle physics, one generally begins with a fixed flat spacetime $\mathbb{R}^4$ and defines fields upon that background. Gravity can then be treated as a field on this background, rather than as the geometry of spacetime itself. This approach is sound as a field-theoretic formulation and provides a natural route toward quantizing gravity and describing gravitons, but it is ugly and inelegant. Regardless, you can bootstrap yourself from there and arrive on the EFE

The distinction of perspectives can be summarized as follows:



Regardless of the formalism, the different approaches must reproduce the same classical predictions. The geometric formulation of GR ultimately leads to the Einstein field equations.

1.1.4 A Theory of Gravity

General Relativity, to be a theory of gravity, must explain gravitational phenomena.

For example, Newtonian gravity became a theory of gravity because it explained the results Kepler had found. Kepler described the motion of the planets and discovered that their orbits were elliptical using data analysis, but he did not have an underlying physical theory explaining *why* they followed these trajectories.

Newton instead derived the observed behavior from first principles. His laws of motion and universal gravitation explained Kepler's laws and provided a general framework for celestial mechanics, and ended up predicting the whole of kinematics and dynamics.

As such, GR as a theory of gravity must be able to explain observed gravitational phenomena: relativistic corrections to celestial mechanics, gravitational waves, black holes, gravitational collapse, and, on the largest scales, cosmology.

Perhaps the grandest application of GR is cosmology. Newtonian gravity is fundamentally a local theory applied within a fixed space and time, while GR allows the spacetime itself to be dynamical. This makes it possible to formulate a theory describing the universe as a whole.

DISTINCTION: For something to be a theory of gravity, what must it possess?

1. **Dynamical variables**

   The theory must have variables describing the gravitational field in which objects move.

$$\begin{aligned} \text{Newton:} &\quad \Phi \\ \text{GR:} &\quad g_{\mu\nu} \end{aligned}$$

2. **A law describing how the field affects objects**

   The theory must describe how its dynamical variables determine the motion of objects.

   For Newtonian gravity:

$$\mathbf{a} = -\nabla\Phi. \tag{1.1.29}$$

   For GR, freely falling objects follow geodesics:

$$\frac{d^2x^\sigma}{d\lambda^2} + \Gamma^\sigma_{\mu\nu}\frac{dx^\mu}{d\lambda}\frac{dx^\nu}{d\lambda} = 0, \tag{1.1.30}$$

   or, using the four-velocity u^μ,

$$u^\mu \nabla_\mu u^\nu = 0. \tag{1.1.31}$$

3. **A law describing how objects produce the dynamical variable**

   Finally, the theory must tell us how matter produces the gravitational field.

   For Newtonian gravity:

$$\nabla^2\Phi = 4\pi G\rho. \tag{1.1.32}$$

   For GR:

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 8\pi G T_{\mu\nu}. \tag{1.1.33}$$

   Since the two formalisms attempt to explain the same phenomenon, we demand that the GR formulation reduces to the Newtonian formulation in weak fields.

Remark :

Notice that $\nabla^2\Phi = 4\pi G\rho$ and $R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 8\pi GT_{\mu\nu}$ are inherently similar. They basically describe the same thing, but in different fields of geometry. Note the following:

$$\underbrace{\nabla^2\Phi}_{\text{some combination of } \partial^2} = \underbrace{4\pi G\rho}_{\text{constant multiplied to matter distribution}}$$

$$\underbrace{R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R}_{\text{some combination of } \partial^2} = \underbrace{8\pi GT_{\mu\nu}}_{\text{constant multiplied by matter distribution}}$$

These three ingredients are the basic structure of a theory of gravity: a dynamical variable, a law governing how it affects matter, and a field equation governing how matter produces it.

GR is inherently nonlinear. Going from the field to the motion of objects is already more complicated than in Newtonian gravity, but the field equation itself is also nonlinear: the metric determines the curvature, while the curvature determines the metric.

Solving the Einstein field equations for realistic systems is therefore often extremely difficult. In many cases, exact solutions cannot be obtained, and the equations must be solved numerically. Careers have been made doing this. This is the domain of *numerical relativity*.

In the next chapters, we will explore how gravity can be explained by general relativity.

1.2 Einstein's Equivalence Principle

1.2.1 Inertial Frames

The Weinberg definition is that, at every point in spacetime, i.e. at every (t, x, y, z) in a gravitational field, it is possible to choose a locally inertial frame, i.e. a freely falling coordinate system, such that the "laws of nature" take the same form as they do in an unaccelerated Cartesian coordinate system in the absence of gravity.

In other words, fields that are accelerating are locally indistinguishable from gravity. This is the basic idea behind the equivalence principle.

Gravity can therefore be regarded as a fictitious force, because it can be removed by a coordinate transformation. It is there only because we are using the wrong coordinate system.

The equivalence principle states that it is always possible to remove gravity locally. Therefore, gravity is a fictitious force" that can be "removed" when describing the local field in an experiment.

The important difference between gravity and other fictitious forces is that gravity can only be removed locally, while other fictitious forces can be removed globally. The reason is that a gravitational field has spatial variation. Even after transforming away the gravitational acceleration at one point, neighboring points may experience slightly different accelerations.

We say that, regardless of the mass of the object, it does not take part in determining its own gravitational acceleration. The gravitational acceleration is universal.

The measurement of differences between these gravitational fields is called *tidal* measurement, or simply the measurement of tides. These tidal effects cannot be transformed away over an extended region.

1.2.2 Universality of Free Fall

The usual way this is expressed is to imagine a person in a closed box with no reference to the outside world. To gain information about his environment, he begins conducting a free-fall experiment. Suppose they perform some experiment in which they can measure the acceleration of objects.

For an object o_1, they measure

$$\mathbf{a}_1 = \frac{m_{1g}}{m_{1i}}\mathbf{g} \tag{1.2.1}$$

and for a second object o_2, they can measure

$$\mathbf{a}_2 = \frac{m_{2g}}{m_{2i}}\mathbf{g}. \tag{1.2.2}$$

The universality of free fall is the statement that

$$\mathbf{a}_1 = \mathbf{a}_2 \tag{1.2.3}$$

under the same circumstances, regardless of the nature or mass of the objects. This is called the *universality of free fall.*

Suppose that the box is accelerating upward with acceleration $g\hat{\mathbf{z}}$.

$$\mathbf{a}_{\text{box}} = g\hat{\mathbf{z}} \tag{1.2.4}$$

The observer releases the ball. From his perspective, the ball appears to accelerate downwards.

$$\mathbf{a}_{\text{ball}} = -g\hat{\mathbf{z}} \tag{1.2.5}$$

Therefore, it is reasonable to conclude that "there is a gravitational field pulling the force downward".

But there isn't. The ball is actually just continuing along its inertial trajectory while the box accelerates upward.

Now compare this with another box which is stationary, with another observer sitting on the surface of Earth.

The observer releases the same ball, and it falls downward with approximately

$$\mathbf{a}_{\text{ball}} = -g\hat{\mathbf{z}} \tag{1.2.6}$$

Inside the box, the experiment looks the same. So locally, motion due to acceleration is indistiguishable from motion incduced by a gravitational field. There is no experiment that can be performed by the two observers that can differentiate one from another. This is the strong equivalence principle.

In Newtonian language, this corresponds to the equality of gravitational and inertial mass. The important point for the equivalence principle is that all sufficiently small freely falling objects respond to the gravitational field in the same way.

Now, imagine the observer is in a box or elevator that is freely falling towards Earth. The box and everything inside it are accelerating downward together. Suppose the observer release a ball inside the box. The ball is falling toward Earth, but so is the box. Since both have essentially the same gravitational acceleration,

$$\mathbf{a}_{\text{ball}} \approx \mathbf{a}_{\text{box}} \tag{1.2.7}$$

their relative acceleration is approximately zero:

$$\mathbf{a}_{\text{relative}} = \mathbf{a}_{\text{ball}} - \mathbf{a}_{\text{box}} \approx 0 \tag{1.2.8}$$

so the ball appears to float. The observer could throw it, and it would move approximately in a straight line at constant velocity relative to the observer.

The observer would therefore reasonably conclude "There is no gravity here. I'm in an inertial frame."

But the observer actually is in Earth's gravitational field. In actual inertial motion, far from any gravitating bodies, the same observations will take place. Therefore, a falling observer is indistinguishable from an inertial observer. That is the weak equivalence principle.

In a simple example, we can write the gravitational field as a non-inertial frame:

$$\ddot{y} = -g \tag{1.2.9}$$

We can remove gravity, however, via certain transformations. Let

$$z = y - \frac{1}{2}gt^2 \tag{1.2.10}$$

This results in

$$\ddot{z} = 0. \tag{1.2.11}$$

This is an inertial coordinate system, since there is no perceived acceleration. The point is that gravity can be removed locally by choosing an appropriate freely falling coordinate system.

The person in the box cannot tell whether they are accelerating upward with constant acceleration g, or whether they are freely falling under the influence of gravity. In a sufficiently small region, the two situations are experimentally indistinguishable.

Gravity is therefore, locally, simply the manifestation of being in the wrong frame.

Inertial means the absence of gravitational acceleration in the local frame. More precisely, an inertial frame is one in which freely falling objects move without coordinate acceleration.

Weak, Strong, and Einstein Equivalence Principles We have the weak equivalence principle, where the "laws of nature" are understood primarily as the laws of mechanics and the universality of free fall. For the strong equivalence principle, one extends the statement to gravitationally self-interacting or strongly gravitating systems, so that the equivalence applies more broadly than to test particles alone.

The Einstein equivalence principle goes further: "laws of nature" genuinely means all laws of nature. It is one of the conceptual foundations of general relativity.

Theorem 1.2 : The Principle of Relativity

The laws of nature take the same form in all inertial observers.

In any theory of gravity, one can attempt to remove g locally by an appropriate choice of coordinates. What makes general relativity special is the geometric interpretation of what remains after the local gravitational acceleration has been removed.

It means that gravity is genuinely fictitious locally if no experiment confined to that sufficiently small region can tell that we are in a gravitational field. If one can remove gravity, then what remains should be the stage itself: the underlying spacetime structure and its geometry.

In summary, over a sufficiently small region of space and time, a fictitious gravitational field produced by acceleration is indistinguishable from a gravitational field produced by mass (strong equivalence principle). Likewise, within a sufficiently small region of space, a freely falling observer cannot distinguish between free fall in a gravitational field and inertial motion in the absence of gravity (weak equivalence principle). Therefore, a freely falling observer is locally an inertial observer.

1.3 History of Relativity

The notion of inertiality plays a major role in relativity. Let us first explain relativity in its historical context.

We begin with special relativity and Newtonian gravity. The combination of these ideas eventually leads us toward general relativity, where gravity is incorporated into the structure of spacetime itself.

1.3.1 Relativity before Einstein

What is relativity? It is an old notion that existed even by the time of Newton. The essential idea is that there should exist observers for which the laws of nature take the same form. Regardless of where the observer is or how they move within the appropriate class of frames, the laws of nature should be invariant.

This is the equivalence of *inertial observers*: the laws of nature take the same form for all inertial observers.

Aristotle did not have this notion. He believed that there was a special place toward which objects naturally moved: the center of the universe, identified in practice with the center of the Earth. One could therefore imagine that this special region defined the natural frame in which the laws of nature took their proper form.

It made sense within that picture to express physics relative to the center of the Earth, since all other frames could be considered as moving relative to this preferred location. For example, one could imagine waves in a pond as being fixed relative to the pond itself, with the pond providing a preferred rest frame.

Galileo rejected this notion by positing that there is not a single preferred inertial state. Instead, there is a class of observers for whom the laws of mechanics are equally valid. These observers move uniformly relative to one another.

This is called *Galilean relativity.* There is a class of observers which all move relative to each other with uniform velocities.

The transformation between two such frames can be written as

$$t' = t \tag{1.3.1}$$

$$\mathbf{x}' = \mathbf{x} - \mathbf{v}t \tag{1.3.2}$$

This relation explains how one can switch to the frame of reference of a second observer moving at velocity $\mathbf{v}$ relative to the first.

In that case, one has the following invariance of Newtonian mechanics:

$$\mathbf{F} = m\mathbf{a} \implies \mathbf{F}' = m\mathbf{a}'. \tag{1.3.3}$$

Newtonian theory therefore satisfies Galilean relativity: it is invariant under Galilean transformations.

From the development of Newtonian mechanics onward, classical mechanics became the fundamental framework of physics. From 1687 onwards, the only fundamental laws of physics is Newtonian physics.

1.3.2 Maxwell's Results

For roughly three centuries, Newtonian mechanics was the jewel of physics. Then electromagnetism broke this picture.

In 1862, Maxwell showed that Maxwell's equations are not Galilean invariant. They therefore do not satisfy Galilean relativity in the same way that Newton's equations do. This challenged centuries of orthodoxy.

The response of academia was initially to deny that Maxwell's equations were fundamental in the same sense as Newtonian mechanics. One possibility was that Maxwell's equations were only correct in the rest frame of some physical medium, and that medium was called the *aether.*

The aether was imagined as an invented medium through which electromagnetic waves propagated. This was reasonable at the time, since all known waves needed a medium to travel from one place to another, think of sound or earthqakes or water waves. In this picture, Maxwell's equations described electromagnetic waves in the same broad conceptual sense that waves in a pond are described relative to the pond.

Light, therefore, would then only move at speed c only in the rest frame of the aether.

This led naturally to attempts to detect the Earth's motion through the aether, such as the Michelson-Morley experiment. However, the results of the experiment was null; the expected difference in the measured speed of light was not observed, and the result indicated that c was the same in all different experimental directions.

People fought against this result, as they should have. The natural response to an experimental discrepancy is to try to preserve a successful theory if possible and search for a physical explanation for the new result.

Ad Hoc Solutions to the Aether Problem. One proposed solution was Lorentz contraction. Objects would contract in the direction of motion through the aether, although there was initially no satisfactory physical explanation for why they should do so. It was essentially an ad hoc solution introduced to maintain consistency with the null result.

Another proposed idea was *ether dragging.* Imagine that the ether is not a fixed frame, but behaves something like molasses or syrup, so that the Earth in motion drags the ether around it. The Earth would then remain locally at rest with respect to the ether, and there would consequently be no observed change in c.

The debates went on for decades. People had built careers around various versions of these theories, and many ad hoc solutions were introduced to preserve the compatibility of Maxwell's theory with Galilean relativity.

There were dozens of theories introduced in the hope of making Maxwell's equations consistent with the Galilean framework.

1.3.3 Einstein's Contribution

Einstein came in 1905 with his *annus mirabilis* papers.

The three major papers associated with this period gave rise to major developments in quantum theory, statistical mechanics, and special relativity. Einstein developed these ideas all before turning 24.

Everyone had been focused on finding a way to mesh Galilean relativity with Maxwell's equations. The common strategy was to deny the fundamentality of Maxwell's equations by claiming that they only worked in the preferred frame of the aether.

Einstein instead reversed the argument.

Einstein posited that Maxwell's equations are fundamental, while Newtonian mechanics is not fundamental. Newtonian mechanics was a theory that had been successful for three centuries, but its enormous historical success did not make it fundamentally immune to revision.

His solution is:

- adopt c as universal;
- accept Maxwell's equations as fundamental.

As such, the equations governing Galilean relativity must be dropped so that c remains constant. By consequence, one also has to give up or modify Newtonian mechanics and stop treating it as fundamentally exact.

One has to divorce oneself from the notion that Newtonian mechanics is fundamental.

The notion of universal observers is still true: there is a class of observers for whom the laws of motion retain the same form. However, to reconcile this with the invariance of c, one must change the transformation law between observers.

This leads to the unification of space and time and to the Lorentz transformations.

Lorentz Transformations and Covariance Maxwell's theory is relativistic, meaning that is relativistically covariant: under the appropriate relativistic transformations, its mathematical form does not change. This covariance is not apparent in the separate electric and magnetic fields, which transform into one another between inertial frames. Relativity instead unifies them as aspects of a single electromagnetic field.

The task is therefore to find the correct transformation laws that keep Maxwell's equations invariant. These transformations will replace the Galilean transformation laws and reveal the kinematic structure required by electromagnetism.

It is then found that the Poincaré-Lorentz transformations, initially formulated to explain the null result of Michelson-Morley and to save the aether hypothesis, provide the appropriate transformations:

Theorem 1.3 : Lorentz Transformations

Transformations of coordinates that allow c to be invariant across all inertial observers.

$$ct' = \gamma(ct - \beta x) \tag{1.3.4}$$

$$x' = \gamma(x - \beta ct) \tag{1.3.5}$$

$$\gamma = \frac{1}{\sqrt{1-\beta^2}} \tag{1.3.6}$$

$$\beta = \frac{v}{c}. \tag{1.3.7}$$

The project, according to Einstein, is to rewrite all of physics so that it is in "relativistically covariant" form: the equations should be invariant in form from one inertial observer to another[3].

The form of these transformations is

$$x^{\mu'} = \Lambda^\mu{}_\nu x^\nu. \tag{1.3.8}$$

where

$$x^\mu = (ct, x^1, x^2, x^3) \tag{1.3.9}$$

and

$$\Lambda^\mu{}_\nu = \begin{bmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{1.3.10}$$

Now, the force equation becomes

$$\frac{d\mathbf{p}}{dt} = \mathbf{F} \implies \frac{dp^\mu}{d\tau} = F^\mu. \tag{1.3.11}$$

The second equation (the four momentum and the four force) reduces to the first equation (the Newtonian vector form), in the appropriate nonrelativistic, low-velocity limit.

[3]This isn't too difficult back in 1905, since the only fundamental theories of physics at the time were Newton's and Maxwells.

How About Gravity? Gravity is simply

$$\nabla^2 \Phi = 4\pi G \rho. \tag{1.3.12}$$

This is not relativistically covariant, since it is written as a theory involving only space and does not incorporate space and time into a unified relativistic structure.

One can try to modify it so that

$$\Box \Phi = 4\pi G T^{\mu}{}_{\nu}\,. \tag{1.3.13}$$

This is known as Nordström gravity. But this theory is incorrect as a theory of gravity because it does not predict the bending of light.

However, Einstein was bothered by the equivalence principle. Why is there an equivalence of falling? Why does gravity have this peculiar property that all objects fall in the same way?

This question leads to the deeper geometric interpretation of gravity.

Minkowski Spacetime It was Minkowski in 1907 who placed these transformations in their modern geometric form.

One can make sense of special relativity by imagining space and time as a single structure: *spacetime.* This spacetime possesses a metric, now known as the Minkowski metric,

$$\eta_{\mu\nu} = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{1.3.14}$$

and the transformations that preserve the Minkowski metric are the correct relativistic transformations.

The key conceptual step is that space and time are no longer treated as two completely separate structures. They form a single spacetime geometry, and inertial observers correspond to the natural geometric structure of this spacetime.

1.3.4 From Special to General Relativity

In 1915, Einstein formalized his theory of General Relativity, where he formalized the notion of spacetime and recognized that spacetime need not be described only by the Minkowski metric.

The replacement is

$$\eta_{\mu\nu} \to g_{\mu\nu}. \tag{1.3.15}$$

The Minkowski metric is therefore promoted from the fixed metric of special relativity to the more general spacetime metric $g_{\mu\nu}$, which may vary from place to place.

Einstein expanded the notion of inertial observers to all observers in free fall within spacetime. Inertial observers are therefore no longer merely observers moving with uniform velocity relative to one another. They are also freely falling observers, following geodesics of the spacetime geometry.

He got the basic idea relatively early, but obtaining the field equations took much longer.

Relativity is very, very old, and the important thing Einstein did was expand it into a general theory in which the geometric structure itself becomes dynamical.

Minkowski's relativity is special relativity, represented by η, called special because it is restricted to a class of observers moving uniformly relative to one another.

The general form, represented by g, becomes general relativity, with gravity incorporated into the geometry of spacetime.

Question: Wasn't Einstein a Machian?

There is a distinction here involving the idea of being *Machian*. A Machian picture suggests that everything should be relational and that nothing should be absolute. Einstein was a Machian, but the theory that he founded is fundamentally not Machian. General relativity has a deeply relational character, although it is not simply Machian in every possible sense. The geometry of spacetime itself becomes part of the physical description rather than serving merely as an immutable background stage.

Relativity is more general than Light There is a common tendency to regard light as being intrinsically tied to relativity. Historically, however, light played a more specific role. Even if Maxwell's theory were somehow removed or replaced by another theory, the principle of relativity would remain.[4]

Light has no fundamental role in the concept of relativity itself. Its importance is primarily historical: Maxwell's equations revealed that electromagnetic waves propagate at a universal speed that is incompatible with Galilean kinematics. This compelled physicists to reconsider the Galilean transformation and ultimately led to the Lorentz transformation. The principle of relativity, however, is conceptually distinct from electromagnetism.

Relativity therefore is a metaprinciple. It is a more general principle concerning the relationship between physical laws and different observers. Maxwell's theory inspired and helped reveal the structure of spacetime, but it is not the conceptual foundation of the principle of relativity itself.

Light is therefore not intrinsically tied to relativity. The relativistic structure of spacetime can be developed without taking electromagnetism or light as its starting point. Likewise, general relativity can be formulated without making light fundamental to its derivation, while still arriving at the same gravitational field equations.

The next lecture will take precisely this approach: we will develop relativity without making any mention of light and arrive at the same equations that describe gravity in Einstein's theory.

The Role of c The essential role of Maxwell's theory in the historical development of relativity is to identify a universal invariant speed, c, that characterizes the structure of spacetime. However, the existence of such an invariant speed does not depend specifically on electromagnetism. We can construct relativistic kinematics with any finite, nonzero value of the invariant speed.

Consider the limiting cases. If we take

$$c \to \infty,$$

[4]Sir here goes on a rant about news sites which sensationalize 'slowing down light speed' experiments as 'putting to question relativity'

we recover Galilean relativity.[5] In this limit, the effects associated with the relativity of simultaneity vanish, and time becomes absolute.

If instead we take

$$c \to 0,$$

we obtain Carrollian relativity,[6] the ultra-relativistic counterpart of the Galilean limit. In this limit, the causal structure becomes degenerate in such a way that spatial motion relative to an inertial observer is effectively frozen.

If instead we asign c to be any fixed finite value,

$$c \in (0, \infty),$$

we obtain the Lorentzian kinematic structure underlying special relativity and, ultimately, general relativity. The particular numerical value of c is therefore not essential to the structure of the theory; any fixed finite nonzero value could serve as the invariant speed, with the numerical value depending on the choice of units and on the physical theory being described.

The constant c is commonly called the *speed of light* because electromagnetic radiation propagates at this invariant speed. Historically, the value of c emerged from Maxwell's theory as the propagation speed of electromagnetic waves. Modern physics, however, understands c more fundamentally as a property of spacetime itself: light travels at c because photons are massless, just as gravitational waves propagate at c. Massive particles cannot reach c, although they may approach it arbitrarily closely.

Thus, Maxwell's theory is not required in order to derive the existence of a finite invariant speed. Rather, it was the first physical theory to reveal that such a speed exists in nature and to give it the numerical value. We can therefore develop a theory of relativity without making any reference to light at all. Starting from the principles of spacetime symmetry, we will see that a finite invariant speed can arise from the kinematic structure itself. We will develop this approach in the next chapter.

[5] This was in fact the case during the time of Galileo. It was a mainstream hypothesis that the speed of light was infinite

[6] This is in reference to Lewis Carrol's *Alice in Wonderland*. It references the Red Queen's race in *Through the Looking-Glass*, where the Queen tells Alice, "Now, here, you see, it takes all the running you can do, to keep in the same place." In Carrollian physics, because $c \to 0$, lightcones collapse entirely, meaning particles cannot move through space no matter how much momentum they have.

Chapter 2

Special Relativity

2.1 Relativity without Light

We begin this section by doing a derivation of special relativity in large contrast with standard methods which introduce relatvity with light-based thought experiments. Our derivation follows the elegant logic developed by von Ignatowsky (1910)[1] and later refined by others, showing that the Lorentz transformations follow from fundamental symmetries alone.

2.1.1 The Transformation Problem

Consider two inertial reference frames S and S', where S is taken as a *rest* frame of reference, and S' moves with a constant velocity v with respect to S. We can treat this as a 1+1 dimensional problem, referring to the position x as the direction where v travels.

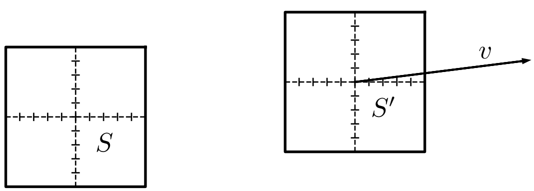


Figure 1: Frames S and S'

In Galilean transformations, the relationship betwen the two reference frames can be described as

$$x' = x + vt$$
$$t' = t'$$

That is, expressed in matrix form,

$$\begin{bmatrix} x' \\ t' \end{bmatrix} = \begin{bmatrix} 1 & v \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ t \end{bmatrix} \tag{2.1.1}$$

This relationship is what is naively assumed during day to day life, and in fact, it is readily apparent for observers without any knowledge of relativity. However, it has some issues. It implies that time t is absolute, and, the reference frames S and S' are not strictly equivalent.

As such, we need a more robust transformation. We seek the most general transformation between the coordinates of the two frames.

$$x' = \overline{X}(x, t; v) \tag{2.1.2}$$
$$t' = \overline{T}(x, t; v) \tag{2.1.3}$$

The functions $\overline{X}$ and $\overline{T}$ are transformation functions that are initially unknown and undetermined.

Our task, then, is to determine them from general physical principles. Simply logic and mathematics, without any reference to experimental data whatsoever. This is, essentially, a purely mathematical approach.

Eventually, our goal is to obtain a transformation in the form of Lorentzian transfromations.

$$\begin{bmatrix} x' \\ t' \end{bmatrix} = \gamma \begin{bmatrix} 1 & -v \\ -\beta & 1 \end{bmatrix} \begin{bmatrix} x \\ t \end{bmatrix} \tag{2.1.4}$$

where $\gamma = \dfrac{1}{\sqrt{1-\beta^2}}$ and $\beta = \dfrac{v^2}{c^2}$

An interesting question arises: Where does c come from if we never assume light?

2.1.2 Principle of Relativity

The first assumption we make is Theorem 1.2, the principle of relativity.

> The laws of nature, that is, the laws of physics, take the same form in all inertial reference frames.

In particular, S and S' are physically equivalent. There is no reason to regard the transformation from S to S' as fundamentally different from the transformation from S' back to S.

If S' moves at velocity v relative to S, then S moves at a velocity $-v$ relative to S'. Hence, the inverse transformation of (2.1.3) must have the same functional form:

$$\begin{aligned} x' &= \overline{X}(x, t; -v) \\ t' &= \overline{T}(x, t; -v) \end{aligned}$$

This implies that if we apply the forward transformation followed by the inverse transformation, we recover the original coordinates:

$$\begin{aligned} x &= \overline{X}\big(\overline{X}(x, t, v), \overline{T}(x, t, v), -v\big) \\ t &= \overline{T}\big(\overline{X}(x, t, v), \overline{T}(x, t, v), -v\big) \end{aligned} \tag{2.1.5}$$

2.1.3 Isotropy of Space

The isotropy of space means that physics has no preferred direction. In particular, the physics cannot distinguish between the $+x$ and $-x$ directions. This imposes the following:

Consider reversing the spatial axis,

$$x \to -x. \tag{2.1.6}$$

Under this reversal, the relative velocity also changes sign,

$$v \to -v. \tag{2.1.7}$$

The spatial coordinate must therefore transform as

$$x' = -\overline{X}(x, t; v), \tag{2.1.8}$$

while time, being a scalar under spatial reflection, satisfies

$$t' = \overline{T}(x, t; v). \tag{2.1.9}$$

This distinction is important: x changes sign under spatial reflection, but t does not.

Under time reversal combined with spatial reflection:

$$\begin{aligned} x' &= -\overline{X}(-x, t, -v) \\ t' &= \overline{T}(-x, t, -v) \end{aligned} \tag{2.1.10}$$

For these to be consistent with the original transformations, we require:

$$\begin{aligned} \overline{X}(-x, t, -v) &= -\overline{X}(x, t, v) \\ \overline{T}(-x, t, -v) &= \overline{T}(x, t, v) \end{aligned} \tag{2.1.11}$$

2.1.4 The Invariance of Straight Lines

Free particles must move along straight lines in both frames. This is required by Newton's first law, the principle of inertia. In space time coodinates, we denote $x^\mu = (x, t)$[7] and $x'^\mu = (x', t')$. Straight line motion in S is therefore characterized by:

$$\frac{d^2 x^\mu}{d\tau^2} = 0 \tag{2.1.12}$$

Where τ is some affine parameter[8].

The principle of inertia demands that straight lines map to straight lines.

$$\frac{d^2 x^\mu}{d\tau^2} = 0 \implies \frac{d^2 x'^\mu}{d\tau^2} = 0 \tag{2.1.13}$$

[7] This is tensor notation, where μ identifies the component. In this 1+1 dimensional case, $x^1 = x$ and $x^0 = t$. However, in general, $x^\mu = \left[t, x, y, z\right]^\top$

[8] An affine parameter is an arbitrary parameter for which the geodesic equation is said to hold and takes the form of uniform motion without introducing artificial acceleration through the choice of parameter. i.e, affine parameters allow reparametrization to be linear transformations. I don't understand it either, go figure. In this case, τ is called the *proper time*

The transformation from S to S' is represented in tensor notation as $x'^{\mu} = \overline{X}^{\mu}(x^{\nu}; v)$. Using the chain rule, with $x^{\nu} = x^{\nu}(\tau)$, we have:

$$\frac{dx'^{\mu}}{d\tau} = \sum_{\nu} \frac{\partial \overline{X}^{\mu}}{\partial x^{\nu}} \frac{dx^{\nu}}{d\tau} \tag{2.1.14}$$

Here, we impose the Einstein summation convention, where twice repeated indices are implicitly summed. Explicitly:

$$\frac{dx'^{\mu}}{d\tau} = \frac{\partial \overline{X}^{\mu}}{\partial x^{0}} \frac{dx^{0}}{d\tau} + \frac{\partial \overline{X}^{\mu}}{\partial x^{1}} \frac{dx^{1}}{d\tau} \tag{2.1.15}$$

Henceforth, we write this compactly as:

$$\frac{dx'^{\mu}}{d\tau} = \frac{\partial \overline{X}^{\mu}}{\partial x^{\nu}} \frac{dx^{\nu}}{d\tau} \tag{2.1.16}$$

with the understanding that ν is summed over its range.

Continuing the derivation, we have

$$\frac{d^2 x'^{\mu}}{d\tau^2} = \frac{d}{d\tau}\left(\frac{\partial \overline{X}^{\mu}}{\partial x^{\nu}}\right) \frac{dx^{\nu}}{d\tau} + \frac{\partial \overline{X}^{\mu}}{\partial x^{\nu}} \frac{d^2 x^{\nu}}{d\tau^2} \tag{2.1.17}$$

The second term vanishes by the straight-line condition in S

$$\cancel{\frac{d^2 x'^{\mu}}{d\tau^2}} = \frac{d}{d\tau}\left(\frac{\partial \overline{X}^{\mu}}{\partial x^{\nu}}\right) \frac{dx^{\nu}}{d\tau} + \cancel{\frac{\partial \overline{X}^{\mu}}{\partial x^{\nu}} \frac{d^2 x^{\nu}}{d\tau^2}} \tag{2.1.18}$$

Therefore, from this result, we demand that the following relation must hold.

$$0 = \frac{\partial^2 \overline{X}^{\mu}}{\partial x^{\nu}\, \partial x^{\sigma}} \frac{dx^{\nu}}{d\tau} \frac{dx^{\sigma}}{d\tau} \tag{2.1.19}$$

For arbitrary velocities $\dfrac{dx^{\nu}}{d\tau}$,the quadratic form above must vanish for every possible velocity vector. this requires:

$$\frac{\partial^2 \overline{X}^{\mu}}{\partial x^{\nu}\, \partial x^{\sigma}} = 0 \tag{2.1.20}$$

Therefore, $\overline{X}^{\mu}$ must be a linear function of the coordinates. Hence, we can write its components as:

$$\overline{X}(x, t, v) = A(v)x + B(v)t + E(v)$$
$$\overline{T}(x, t, v) = C(v)x + D(v)t + F(v)$$

These are just general equations of straight lines.

2.1.5 Homogeneity of Spacetime

The next assumption we use is the homogeneity of space time. Spacetime homogeniety implies no preferred origin.

Any point in both S and S' can be an origin and spacetime would remain the same, so therefore we can choose the location of the origin for both.

Therefore, we demand that the origin of S gets mapped to the origin of S' at $t = 0$.

$$x = t = 0 \iff x' = t' = 0 \tag{2.1.21}$$

This is not really essential. However, it makes the derivation simpler and easier, since we can then set the constants $E(v)$ and $F(v)$ to zero. Doing otherwise would make the constant terms indeterminable, which would increase the difficulty of the problem. However, the general result is the same. With this assumption, we can reduce our transformations to

$$\begin{aligned} \overline{X}(x,t,v) &= A(v)x + B(v)t \\ \overline{T}(x,t,v) &= C(v)x + D(v)t \end{aligned} \tag{2.1.22}$$

Thus, only four free functions of velocity remain to be determined: $A(v), B(v), C(v)$, and $D(v)$. Our task, then, is to reduce these four free functions to just one function by exploiting properties that were previosuly shown.

2.1.6 Parity Constraints

Applying the parity conditions from Eq. (2.1.11) to the linear forms, we have

$$\begin{aligned} \overline{X}(-x,t,-v) &= -\overline{X}(x,t,v) \\ \overline{T}(-x,t,-v) &= \overline{T}(x,t,v) \end{aligned} \tag{2.1.23}$$

This yields the following:

$$\begin{aligned} -A(-v)x + B(-v)t &= -A(v)x - B(v)t \\ -C(-v)x + D(-v)t &= C(v)x + D(v)t \end{aligned} \tag{2.1.24}$$

For these to hold for all x and t, the following must be true

$$\begin{aligned} A(-v) &= A(v) \\ D(-v) &= D(v) \\ B(-v) &= -B(v) \\ C(-v) &= -C(v) \end{aligned} \tag{2.1.25}$$

Thus, $A(v)$, $D(v)$ are even functions of v, and $B(v)$, $C(v)$ are odd functions of v

2.1.7 Inverse transformation and composition constraints

The inverse transformation from S' to S is

$$\begin{bmatrix} x \\ t \end{bmatrix} = \begin{bmatrix} A(-v) & B(-v) \\ C(-v) & D(-v) \end{bmatrix} \begin{bmatrix} x' \\ t' \end{bmatrix} \tag{2.1.26}$$

Explicitly,

$$\begin{aligned} x &= A(-v)x' + B(-v)t' \\ t &= C(-v)x' + D(-v)t' \end{aligned} \tag{2.1.27}$$

Using the parity relations (2.1.25),

$$\begin{aligned} x &= A(v)x' - B(v)t' \\ t &= -C(v)x' + D(v)t' \end{aligned}$$

Substituting

$$\begin{aligned} x' &= A(v)x + B(v)t \\ t' &= C(v)x + D(v)t \end{aligned}$$

to (2.1.27), we obtain

$$\begin{aligned} x &= A(Ax + Bt) - B(Cx + Dt) \\ t &= -C(Ax + Bt) + D(Cx + Dt) \end{aligned}$$

Collecting coefficients we have,

$$\begin{aligned} x &= (A^2 - BC)x + B(A - D)t \\ t &= C(D - A)x + (D^2 - BC)t \end{aligned}$$

Since x and t are arbitrary, the coefficients must satisfy

$$\begin{aligned} A^2 - BC &= 1 \\ B(A - D) &= 0 \\ C(D - A) &= 0 \\ D^2 - BC &= 1 \end{aligned} \tag{2.1.28}$$

So far, nothing specifically relativistic has appeared, there's no magic yet. We just used natural assumptions that come from geometry, and principle of relativity.

2.1.8 The Trivial and Nontrivial Branches

The conditions

$$B(A - D) = 0 \tag{2.1.29}$$

$$C(D - A) = 0 \tag{2.1.30}$$

give us two cases.

Case 1: $A \neq D$

If $A \neq D$, then, we are forced to conclude

$$B = C = 0 \tag{2.1.31}$$

The remaining equations give

$$A^2 = D^2 = 1 \tag{2.1.32}$$

Therefore,

$$\begin{aligned} A &= \pm 1 \\ D &= \mp 1 \end{aligned}$$

Thus, the transformation becomes

$$\begin{aligned} x' &= \pm x \\ t' &= \mp t \end{aligned}$$

This is a trivial transformation, corresponding to an identity/reflection type transformation rather than a continuosly varying boost between each reference frame. This is obvious. Of course, when you flip a frame, all the assumptions and constraints still hold. This is uninteresting.

They are therefore not the nontrivial transformation we seek.

Case 2: $A = D$

This is the nontrivial condition. The physically interesting case is therefore

$$A(v) = D(v). \tag{2.1.33}$$

The condition

$$A^2 - BC = 1 \tag{2.1.34}$$

then gives

$$BC = A^2 - 1. \tag{2.1.35}$$

provided that $B \neq 0$ Thus:

$$C = \frac{A(v)^2 - 1}{B(v)}. \tag{2.1.36}$$

We have reduced the four unknown free functions to two, since $A = D$, and once you know A and B, you know C.

2.1.9 The motion of the origin of S'

Consider the motion of S' through S. The spatial origin of (S') is defined by

$$x' = 0 \tag{2.1.37}$$

Since S' moves at velocity v relative to S, its origin follows the trajectory

$$x = vt \tag{2.1.38}$$

But every event on this trajectory must satisfy $x' = 0$. Again, that is for the spatial origin. That means that entire *worldline* needs to be mapped to $x' = 0$.

Using

$$x' = Ax + Bt \tag{2.1.39}$$

we therefore have

$$\begin{aligned} 0 &= Avt + Bt \\ &= (Av + B)t \end{aligned}$$

Since this must hold for an arbitrary t, then

$$B(v) = -vA(v) \tag{2.1.40}$$

Now we only have one free function, $A(v)$.

Consequently, we have

$$C = \frac{A^2(v) - 1}{vA(v)} \tag{2.1.41}$$

Thus, we have reduced the transformation to the following

$$\begin{bmatrix} x' \\ t' \end{bmatrix} = \begin{bmatrix} A & -vA \\ -\dfrac{(A^2-1)}{vA} & A \end{bmatrix} \begin{bmatrix} x \\ t \end{bmatrix} \tag{2.1.42}$$

Now we only need to know A, except we dont know what properties of A is. The only thing we can say for now is that matrix needs to reduce to the identiy matrix when $v = 0$

Thus we impose

$$A(v) \to 1, \quad v \to 0.$$

Since the $M^1{}_0$ term is a fraction with v in the denominator, it must vanish sufficiently fast. therefore, $A^2 - 1 \approx \mathcal{O}(v^2)$

2.1.10 Introduction of a Third Observer

Consider a third inertial frame S''.

Suppose S'' moves with constant velocity u relative to S', while S' moves with constant velocity v relative to S.

It's the same framework, then. The transformation from S' to S'' takes a similar form as from S to S'. Let M be a matrix such that

$$M^\mu{}_\nu(v) = \begin{bmatrix} A(v) & -vA(v) \\ -\dfrac{(A(v)^2-1)}{vA(v)} & A(v) \end{bmatrix} \tag{2.1.43}$$

We can therefore denote the transformation from the S frame to the S' frame to be

$$x'^\mu = M^\mu{}_\nu(v) x^\nu \tag{2.1.44}$$

Since the transformation is the same, then the transformation from S' to S'' is

$$x''^\mu = M^\mu{}_\nu(u) x'^\nu \tag{2.1.45}$$

Suppose that S'' is moving at a speed w relative to S. Then

$$x''^{\mu} = M^{\mu}{}_{\nu}(w)x^{\nu} \tag{2.1.46}$$

However, we demand the closure of transformations under composition. i.e, the form of the transformation is the same, just with a different argument. Then, the transformation from S to S'' is obtained by composing the two transformations from the intermediate reference frame.

$$x''^{\mu} = M^{\mu}{}_{\nu}(u)M^{\nu}{}_{\sigma}(v)x^{\sigma} \tag{2.1.47}$$

That is, the form of the transformation from S to S'' is the same form as the transformations from the other frames of reference.

$$M^{\mu}{}_{\sigma}(w) = M^{\mu}{}_{\nu}(u)M^{\nu}{}_{\sigma}(v) \tag{2.1.48}$$

The crucial requirement is that the composition of two allowed transformations must itself be an allowed transformation.

2.1.11 The Emergence of an Invariant Speed

We evaluate the composition by doing matrix multiplication.

$$M^{\mu}{}_{\nu}(w) = \begin{bmatrix} A(u) & -uA(u) \\ -\dfrac{(A(u)^2-1)}{uA(u)} & A(u) \end{bmatrix} \begin{bmatrix} A(v) & -vA(v) \\ -\dfrac{(A(v)^2-1)}{vA(v)} & A(v) \end{bmatrix}$$

Therefore, we have

$$M^{\mu}{}_{\nu}(w) = \begin{bmatrix} A(u)A(v) + \dfrac{uA(u)\big(A(v)^2-1\big)}{vA(v)} & -(u+v)A(u)A(v) \\ -\dfrac{\big(A(u)^2-1\big)A(v)}{uA(u)} - \dfrac{A(u)\big(A(v)^2-1\big)}{vA(v)} & A(u)A(v) + \dfrac{vA(v)\big(A(u)^2-1\big)}{uA(u)} \end{bmatrix}$$

By the nature of M, we see that the diagonal elements of the transformation matrix are equal, $M^{0}{}_{0} = M^{1}{}_{1}$. Therefore, we require the equality of the elements. We equate:

$$A(u)A(v) + \frac{uA(u)\big(A(v)^2-1\big)}{vA(v)} = A(u)A(v) + \frac{vA(v)\big(A(u)^2-1\big)}{uA(u)} \tag{2.1.49}$$

Simplifying, we find that

$$\cancel{A(u)A(v)} + \frac{uA(u)\big(A(v)^2-1\big)}{vA(v)} = \cancel{A(u)A(v)} + \frac{vA(v)\big(A(u)^2-1\big)}{uA(u)}$$

$$\frac{uA(u)\big(A(v)^2-1\big)}{vA(v)} = \frac{vA(v)\big(A(u)^2-1\big)}{uA(u)} \tag{2.1.50}$$

$$\frac{\big(A(v)^2-1\big)}{v^2A^2(v)} = \frac{\big(A(u)^2-1\big)}{u^2A^2(u)} \tag{2.1.51}$$

The left hand side only depends on u, and the right hand side only depends on v. Since u and v are arbitrary, both sides must equal a universal constant.

Let that constant be k:

$$\frac{A^2(v)-1}{v^2A^2(v)} = k \tag{2.1.52}$$

Rearranging, we have

$$A^2(v) - 1 = kv^2A^2(v)$$
$$A^2(v)(1-kv^2) = 1$$

Therefore,

$$A = \frac{1}{\sqrt{1-kv^2}} \tag{2.1.53}$$

where we choose the positive branch because the transformation must continuously approach the identity as $v \to 0$, thus reducing to the Galilean limit.

$$A(0) = 1 \tag{2.1.54}$$

Note the units of k. For A to be dimensionless, it has to have units which are $[L^{-2}][T^2]$, i.e, the inverse of the units of the square of velocity.

Now, notice that M_{10} reduces to

$$\frac{A^2-1}{vA} = kvA \tag{2.1.55}$$

Thus the transformation becomes

$$\begin{bmatrix} x' \\ t' \end{bmatrix} = A \begin{bmatrix} 1 & -v \\ -kv & 1 \end{bmatrix} \begin{bmatrix} x \\ t \end{bmatrix} \tag{2.1.56}$$

Equivalently,

$$x' = \frac{x - vt}{\sqrt{1 - kv^2}} \tag{2.1.57}$$

$$t' = \frac{t - kvx}{\sqrt{1 - kv^2}} \tag{2.1.58}$$

This are already the Lorentz transformations in generalized form.

From this, we can derive that[9]

$$w = \frac{u + v}{1 + kuv} \tag{2.1.59}$$

This is the familiar relativistic velocity addition formula.

Writing k always in terms of $1/k^2$ is a bit cumbersome, especially when writing longer proofs. We hereby assign a new symbol for the speed. Let $\kappa = \frac{1}{\sqrt{k}}$.

Suppose we set $u = \kappa$. Then, we find

$$w = \frac{\kappa + v}{1 + kv\kappa} \tag{2.1.60}$$

Since $k\kappa = \dfrac{1}{\kappa}$, we then obtain

$$\begin{aligned} w &= \frac{\kappa + v}{1 + \frac{v}{\kappa}} \\ &= \frac{\kappa + v}{(\kappa + v)/\kappa} \\ &= \kappa \end{aligned}$$

thus:

$$u = \kappa \implies w = \kappa \tag{2.1.61}$$

The speed κ is therefore invariant under the velocity-composition law.

This is the crucial result from all that derivation. The principle of relativity does not tell us what the invariant speed is. It tells us that:

Theorem 2.1 : Universal Invariant Speed

For the Lorentzian branch of the kinematics, there must exist a universal invariant speed κ.

[9] To be proven.

2.1.12 The Relevance of Light

Where did light enter the discussion? It has not entered the argument at all. We only began with the following fundamental principles

1. principle of relativity
2. homogeneity of space
3. homogeneity of time
4. isotropy of space
5. principle of inertia
6. closure of transformations under compositions

and these assumptions led us to find the transformation equations. Nowhere did Maxwell or electromagnetics come into the picture.

We did not need light at all. In this sense, the theory of relativity is derived and works on pure mathematical logic alone. All of physics bows down to this principle.

The empirical question that can then be raised is: Which value k describes our universe?

Experiment tells us that the invarient speed is the speed of light in a vacuum.

$$\kappa = c \tag{2.1.62}$$

so that

$$k = \frac{1}{c^2} \tag{2.1.63}$$

It did not need to be light. If we were more sensitive to gravitational waves, we would have called it thus. However, we are a species who evolved to sense light better, and it was this that lead us to stumble upon relativity. But there is in fact no need at all for it. A a blind society, insensitive to light, just with principles and logic, can establish relativity. Thus, electromagnetism does not need to be used to *derive* the Lorentz transformation. Instead, electromagnetism provides us an empirical way of identifying the invarient speed.

In this sense, light acts as a calibrator of the kinematics rather than as the logical foundation of the transformation.

2.1.13 The Galilean Limit

There is another possibility.

If we set $\kappa \to 0$, then $\kappa \to \infty$ and $A \to 1$, the transformation thus becomes

$$\begin{bmatrix} x' \\ t' \end{bmatrix} = \begin{bmatrix} 1 & -v \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ t \end{bmatrix}$$

These are precisely the Galilean transformations. The corresponding velocity addition law is then

$$w = u + v$$

In this case, the invariant speed is effectively infinite. Thus, Galilean relativity can be viwed as the $\kappa \to \infty$ limit of the generalized transformation.

2.1.14 Lorentzian, Galilean, and Carrollian kinematics

The parameter k therefore distinguishes different possible kinematic structures.

For $k > 0$, there is a finite invariant speed

$$\kappa = \frac{1}{\sqrt{k}}$$

giving us Lorentzian relativity.

For $k = 0$, the invariant speed becomes infinite, giving Galilean relativity. The main reason that Galilean relativity took hold for so long is for quite some time in the enlightenment, people thought that the speed of light is infinite, and so that is the logical conclusion.

While inaccurate at relativistic scales, Galilean relativity, and by extension Newtonian physics, remains central to physics education because, for systems involving velocities $v \ll c$ and ordinary spatial and temporal scales, it provides an excellent approximation to relativistic physics. In this regime, the corrections predicted by special relativity are negligible, making the Galilean description both sufficiently accurate and considerably simpler as a description of reality.

There is also a limiting case in which the invariant speed tends to zero. As $k \to \infty$,

$$\kappa \to 0 \tag{2.1.64}$$

This is associated with the Carrolian limit of relativity. In this regime, the causal structure becomes increasingly restrictive: events at distinct spatial locations cannot influence one another within any finite time interval, and the notion of spatial propagation effectively disappears. The resulting Carrollian kinematics therefore describes a theory in which particles may evolve in time while remaining effectively frozen in space, providing the opposite extreme to the Galilean limit of instantaneous propagation.

These three structures can therefore be understood schematically with light cones, representing the travel of information in an observer's frame of reference.

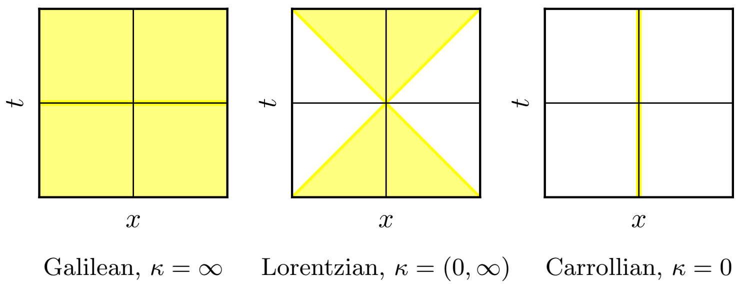


Figure 2: Light cones in different types of relativity

The light cone structure which posits finite travel of information belongs to the finite Lorentzian case. The causal cone has a finite opening angle because there is a finite maximum invariant speed. In the Galilean limit, the cone opens out to incompass instantaneous propagation of information, no matter the distance of the interaction. In the Carollian limit, the cone collapses, corresponding to the extreme opposite situation where spatial propagation, and travel of information, is suppressed, no matter now much momentum a particle carries. As such a body can only travel forwards in time, fixed in the same space, with no information reaching it. Such a universe would be mathematically interesting, but in effect, it is a dead universe.

2.1.15 Conclusion

This, and this is a point that I will not get of tired telling you,[10] proves the point that the derivation of the Lorentz transformation does not logically require the existence of light. The transformation

$$\begin{bmatrix} x' \\ t' \end{bmatrix} = \gamma \begin{bmatrix} 1 & -v \\ -\beta & 1 \end{bmatrix} \begin{bmatrix} x \\ t \end{bmatrix}$$

can be obtained from the structure of inertial frames and the relativity principle alone, with the constant c initially appearing simply as an arbitrary invariant speed. Experiment then tells us that this invariant speed is

$$c = 299\,792\,458\,\mathrm{m/s} \tag{2.1.65}$$

and that electromagnetic wave propagation also happens to have an invariant speed, and that speed turned out to be c as well, for electromagnetic radiation travel precisely this speed in a vacuum.

The logical relationship can be summarized as

$$\text{Relativity} \implies \text{possible invariant speed } \kappa$$

while experiment establishes

$$\kappa = c$$

Electromagnetism therefore provides an empirical realization of the invariant speed; it is not necessary as a premise in the kinematic derivation itself. The deeper lesson is that the Lorentz transformations and relativity itself is fundamentally a statement about the symmetry of spacetime, not specifically about light.

Remark :

Some points because I wasn't able to elaborate on them. $\overline{X}^{\mu}(x^{\nu})$ is simply the tensor packaging of the transformation functions, where $\overline{X}^{0}(x^{\nu};v) = \overline{X}(x,t;v)$ and $\overline{X}^{1}(x^{\nu};v) = \overline{T}(x,t;v)$. It is a vector valued function. Whereas $M^{\mu}{}_{\nu}$ is a matrix. Once linearity is proven, we can show that $M^{\mu}{}_{\nu}$ is a Jacobian, related to $\overline{X}^{\mu}$ by

$$M^{\mu}{}_{\nu} = \frac{\partial \overline{X}^{\mu}}{\partial x^{\nu}}$$

[10] Actual verbatim quote from sir

Addendum

In §2.1.11, we are given the velocity addition formula

$$w = \frac{u+v}{1+kuv}$$

and told it is true without proof, despite being pivotal to the proof that there is an invariant speed that arises in Lorentzian mechanics.

This wasn't elaborated in the lecture for time constraint reasons, as Sir was rushing to finish the invariant speed derivation before class dismissal, and thus it was left as an exercise to the student. Finding no satisfaction in it being simply stated as true, I took the liberty of proving the relation.

Proof. From §2.1.10, we know that the origin of the second frame S'' has velocity

$$w = \frac{dx}{dt} \tag{2.1.66}$$

as measured from S.

The same reference frame has velocity

$$u = \frac{dx'}{dt'}$$

as messured from S'.

We also know that S' travels at a velocity v relative to S. We aim to determine w in terms of u and v.

From the generalized Lorentz transformation, we have the transformed coordinates in S'

$$x' = \frac{x - vt}{\sqrt{1-kv^2}} \qquad t' = \frac{t - kvx}{\sqrt{1-kv^2}}\text{“}$$

Taking the differentials of these equations give

$$dx' = \frac{dx - v\,dt}{\sqrt{1-kv^2}} \qquad dt' = \frac{dt - kv\,dx}{\sqrt{1-kv^2}}$$

Hence, we have

$$\begin{aligned} u &= \frac{dx'}{dx'} \\ &= \frac{dx - v\,dt}{dt - kv\,dx} \end{aligned}$$

We divide both the numerator and denominator by dt, we have

$$u = \frac{\frac{dx'}{dt}}{\frac{dt'}{dt}}$$

$$= \frac{\frac{dx}{dt} - v\frac{dt}{dt}}{\frac{dt}{dt} - kv\frac{dx}{dt}}$$

Since we know that (2.1.66), and $\frac{dt}{dt} = 1$, then we obtain.

$$u = \frac{w - v}{1 - kvw} \tag{2.1.67}$$

Solving for w, we find that

$$\begin{aligned}
u(1 - kvw) &= w - v \\
u - kuvw &= w - v \\
u + v &= w + kuvw \\
u + v &= w(1 + kuv) \\
w &= \frac{u + v}{1 + kuv}
\end{aligned}$$

Therefore, the velocity of S'' relative to S is:

Theorem 2.2 : Relativistic Velocity Addition

$$w = \frac{u + v}{1 + kuv} \tag{2.1.68}$$

$\square$

Chapter 3

Aspects of Differential Geometry

3.1 Manifolds and Topological Spaces

This lecture was conducted online due to weather constraints, and so I had time to pause and rewind the lecture. Thus, most of what is written here is verbatim.

Last meeting, we discussed that relativity functions without the need of light, and derived the equations of SR without any assumptionss pertaining to light and assumptions that were more natural coming from established or logical principles.

What we will learn in subsequent lectures is that SR, at least in the way we understand it now, is really about physics in Minkowski spacetime.

And so, we will discuss what spacetime is today, and one of the foundational concepts in understanding spacetime is the notion of manifolds. So today, we shall start talking about manifolds. This is gonna be a very lightning quick introduction to manifolds.

So why manifolds? Why are they important? Before we get to manifolds, let's first understand what spacetime actually is, and what are topological spaces and other preliminaries.

3.1.1 Spacetime

Spacetime, in an intuitive sense, is just a collection of all *events* in physics. An event is just a *here* and a *now*, so you can think of spacestime as a here and a now. It's very intuitive, the way a philosopher might want to describe it. Spacetime, as we understand it, is just[11] the background on which physics takes place. The stage, so to speak.

It's hard to imagine physics without actually thinking about events. Say, for example, an electromagnetic field, there's always some kind of field vector, like an electric field vector or a magnetic field vector existing at some spatial point and some instant of time. We think about some sort of arrow existing on some kind of background, traditionally $\mathbb{R}^3 + t$, and that's what we think spacetime is. There is always spacetime, there has always been space and time in physics, but for most cases, we didn't give it much thought, right? It was simply a stage at which physics happens, and where particles and fields effectively *play*.

However, modern physics is different. In fact, in some sense, this is a *characteristic* of modern physics, we have to take into account the role played by spacetime because spacetime is not just some inert object anymore, in general relativity in particular, because in fact all modern relativistic theories have to contend with the empirical fact that spacetime does have dynamical role serving a stage for everything else.

Modern physics requires us to be careful of this *stage*, because it participates greatly in how reality behaves. In particular, we have to be careful about the structures which live on this stage.

So what am I talking about here, what *structures*? These are structures that we are all familiar with, such as scalars, vectors, and there will be a spinors sometiems, one forms, but in general, all of these will be subsumed by what we know to be tensors. So all these objects are the objects that can naturally live on space time. We want these to be the things that represent the physical objects that we're used to, such as fields and particles, and we need a mathematical structure which can contain or allow the existence of these objects and this will turn out most naturally to be a manifold.

Before relativity, as I've explained in previous lectures, the implicit assumption that there was some kind of background that was just plainly Euclidean. Pre-relativity, before Einstein's time, there was just the New-

[11] we say just because this was how it was conceived back in the olden times

tonian picture. For example there was already a space time in those times, a space structure and a time structure. Here, it's probably not very appropriate to join space and time, because it was really space *and* time, to have a Euclidean structure, at least a Euclidean structure for space. In fact, it was a very specific kind of space time, it was a manifold[12]

$$\mathcal{M} = \mathbb{R} \times \mathbb{R}^3 \tag{3.1.1}$$

In other words, it had the familiar Euclidean space $\mathbb{R}^3$ for space and time. So you can sort of visualize it as something like this

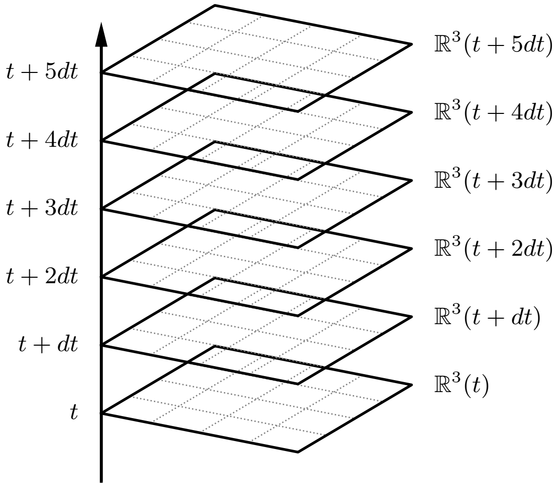


Figure 3: Conception of space time before relativity

So each point in time t (which is a real line $\mathbb{R}$), which is absolute global universal time, there is an associated copy of $\mathbb{R}^3$. Time is going forward, and space is *coupled* to each point in time, and there would be a particle going through this space time. This was the structure of pre-Relativity physics, This was before Minkowksi. Bear in mind that this is not Euclidean $\mathbb{R}^4$, but simply distinct $\mathbb{R}$ and $\mathbb{R}^3$ structures.

Furthermore, for this Euclidean space, there is also an assumed

[12] which is what we generally call what a spacetime is

distance metric, which measures the distances between points in $\mathbb{R}^3$. For Euclidean space, this is basically the distance formula.

$$d(u,v) = \sqrt{\sum_i (u_i - v_i)^2}, \qquad u, v \in \mathbb{R}^n$$

But in the modern conception of spacetime, it has to be far more general, in recognition of the insights of Einstein and many others. So, we'll provide a more modern definition. In fact, there are several definitions of spacetime, they all roughly correspond to the same thing. Some are just more careful because some are more mathematically oriented. But I shall provide two definitions just for your reference. The first one is by Sasane[2], he was a mathematician interested in relativity.

Definition 3.1 : Spacetime (Sasane)

Spacetime, $(\mathcal{M}, g)$ is a connected four dimensional, oriented (time oriented) Lorentzian manifold $\mathcal{M}$ with metric g, equipped with a Levi-Civita connection, induced by g, together with a time orientation.

There are a couple of things we should take note of here. Notably, there are terms like *connected*, which we'll explain in due course. *Four dimensional*, which more or less we have a notion of, and there is a *manifold*, which we're going to talk about today. There is the term *Lorentzian*, which actually refers to the metric g, and there's also a notion of the *Levi-Civita connection*, and somehow this Levi-Civita connection is supposed to come from g, presumably. What this means is that if there is a metric g, it implies a Levi-Civita connection. And we have *time orientation*. All of these terms, we will have to parse in the days ahead.

The more classic definition of spacetime comes from the great Hawking and Ellis[3],[13] The definition they provide is a little bit simpler than Sasane's which is more modern, they say that

[13] If you are particularly interested in understanding relativity from a mathematical point of view, I will highly recommend Hawking and Ellis as a book, but not as a first textbook. There are still aspects of Hawking and Ellis that are a bit difficult for me to understand, but this has been a standard reference for some time.

Definition 3.2 : Spacetime (Hawking and Ellis)

Spacetime is the collection of all events, represented by a pair $\mathcal{M}, g$, where $\mathcal{M}$ is a connected four dimensional Hausdorff C^∞ manifold and g is a Lorentz metric

There are some common elements, such as the terms *connected* and *four dimensional*. There is a topological feature here: *Hausdorffness*. This is a separability assumption, we will discuss this in a bit. And then there is a notion of a C^∞ manifold, which we will discuss today as well, and then there is a Lorentz metric and g, both definitions pertain to the same thing, which we will make sense of in the days ahead.

First we discuss the C^∞ manifold, or manifolds in general.

3.1.2 Manifolds

Those definitions discuss both an $\mathcal{M}$ manifold and a g metric. As such, we discuss what is a manifold. First, I will write down what I consider to be a sufficient definition for what a manifold is, and then I'll give you my own take of how do you make sense of a manifold, from an intuitive point of view.

Definition 3.3 : Manifold

An n-dimensional differentiable or smooth manifold[a] $\mathcal{M}$ is a topological space with open cover[b] $\{\mathcal{O}_\alpha\}$ (i.e., $\mathcal{M} = \cup_\alpha \mathcal{O}_\alpha$) and charts $\{(\psi_\alpha, \mathcal{O}_\alpha)\}$ such that each ψ_α is a homeomorphism and if $\mathcal{O}_\alpha \cap \mathcal{O}_\beta \neq \varnothing$, the composite map $\psi_\beta \circ \psi_\alpha^{-1}$ is C^∞ (or smooth).

[a] In this class when we talk about manifolds we always mean differentiable or smooth manifold

[b] If you take the union of all these open sets then it gives you the whole manifold

The gist is that a differentiable manifold or a smooth manifold is something that is coordinatizable. So from a physicist's standpoint, that's a way to understand what a manifold is. An n-dimensional manifold is a topological space that is locally homeomorphic to $\mathbb{R}^n$, together with appropriate compatibility within charts. It has the local differential structure of $\mathbb{R}^n$ but not necessarily its global properties.

In short: A manifold is basically a topological space on which you can install coordinates in. It is a topological space which is coordinatizable.

So, obviously there are numerous examples when you need coordinatizability and it's very difficult to actually imagine situations in fundamental physics in which there are no coordinates, and there are many examples of these manifolds in physics, in fact manifold structures are implicit for most branches of physics. For those branches of physics you don't need to be very careful about what the manifold is, so long as you're using coordinates to describe your physics. However, if you need to place coordinates in your physics, you need the structures given by manifolds.

For example, Thermodynamics, so when you think about the state spaces there, coordinatized by, say $P_V T$, for instance, thermodynamic variables, those are examples of manifolds. When you're doing mechanics, for example, you deal with phase spaces, phase spaces typically have coordinates that you can configure, such as q_α and p_α, those are also manifolds.

Any application or situation, where you have coordinates, you implicitly use manifolds. The important point is that in GR, or relativity, we have to be careful about the manifold and not just be so focused on the coordinates because the coordinates are just things that live on top of a manifold.

I'm hoping that's clear, but we'll make that even clearer in a bit.

Aside from being locally coordinatizable, it is also always[14] locally $\mathbb{R}^n$.

In other words, if a manifold exists (Fig 4), and you take a small part there, that small part of it will always have a set of n coordinates. That is, a set of n numbers describe the members of the set, that's what it means. Locally $\mathbb{R}^n$ just means that you can always find n numbers to completely describe the elements of any set $\mathcal{O}_\alpha$. But before we get to that, now we talk about topological spaces; this is gonna be another very lightning quick review of what a topology is, because this is not a mathematical methods class, this is really a GR class but we have to say a little bit about it.

[14] i.e., every point has a neighborhood homeomorphic to an open subset of $\mathbb{R}^n$

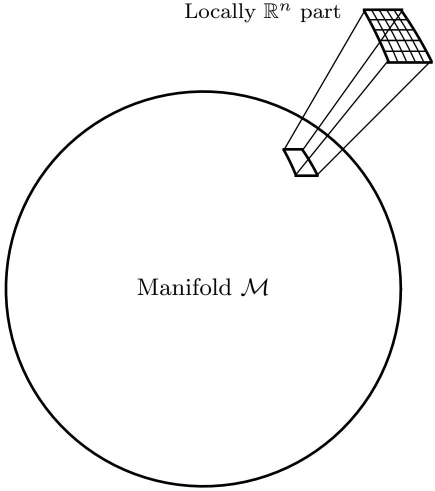


Figure 4: Manifold example

3.1.3 Topological Space

A topological space is just a set with a topology. What is a topology? A topology on a set $\mathcal{M}$ is a collection $\mathcal{T}$ of subsets of $\mathcal{M}$, called open sets, satisfying three axioms defined below. In other words, a topology is an abstract specification of what is called open balls or open sets, usually $\mathbb{R}^n$, denoted by *the set of open sets*, $\{\mathcal{O}_\alpha\}$. If you have a set $\mathcal{M}$, you can identify subsets of that set, and call it open sets. But you can't just name any subset to be an open set, they have to satisfy certain rules or criteria. These criteria are:

(1) The set $\mathcal{M}$ and the null set $\varnothing$ are both open.

$$\mathcal{M}, \varnothing \in \{\mathcal{O}_\alpha\}$$

Since $\{\mathcal{O}_\alpha\}$ is defined as the set of all open sets, we require in the specification of open set that $\mathcal{M}$, the full set, and the empty set $\varnothing$, are both elements of the set of open sets.

(2) If you take an arbitrary union of open sets, it must also be open

$$A_i \in \{\mathcal{O}_\alpha\} \implies \bigcup_i A_i \in \{\mathcal{O}_\alpha\}$$

For example, you already specified the open sets in the set, and you take the any union of any of those things you named in the open set, their union must also be open. It holds regardless of how many unions of sets you take. It must also be an element of $\{\mathcal{O}_\alpha\}$

(3) Finite intersections of open sets must be open.

$$A_1, A_2, \ldots, A_n \in \{\mathcal{O}_\alpha\} \implies \bigcap_{i=1}^{n} A_i \in \{\mathcal{O}_\alpha\}$$

for any positive integer n.

That is, for any open set you define, the finite intersection of how many such sets you take, must also be open. The *finite* restriction in this case is crucial.

That's it, that's all we need to define an open set. I'll give you a very easy example, it's very trivial, it's hardly useful but it's good for understanding definitions, it's what's known as the indiscrete topology.

Indiscrete Topology In the indiscrete topology, we define that the only open sets are $\mathcal{M}$ and the null set $\varnothing$.

$$\{\mathcal{O}_\alpha\} = \{\mathcal{M}, \varnothing\}$$

Is this a topology? Clearly it satisfies the first condition, by construction. The required members of the topology are already open sets by declaration. Does this satisfy the second criteria of arbitrary unions? Well, the only union you can take is between $\mathcal{M}$ and the empty set, and itselves, and as such:

$$\mathcal{M} \cup \varnothing = \mathcal{M} \qquad \mathcal{M} \cup \mathcal{M} = \mathcal{M} \qquad \varnothing \cup \varnothing = \varnothing$$

But $\mathcal{M}$ is already an open set by definition, as is $\varnothing$. Therefore, it satisfies the second criteria. What about finite intersections? Here, it is not really relevant because the open set is already a finite set with only two elements. Similarly, the only intersection we can take is

$$\mathcal{M} \cap \varnothing = \varnothing \qquad \mathcal{M} \cap \mathcal{M} = \mathcal{M} \qquad \varnothing \cap \varnothing = \varnothing$$

and all of these are already members of the set of open sets, so the third criteria is already satisfied. So this is an example, the simplest example, of a topology. You identify subsets, not necessarily proper subsets, but subsets of your set of open sets, and you call them open. It's a declaration of what the open sets in your set are.

This is what's known as the coarsest topology, because it only has two open sets. It is practically useless for application, but again it's good for understanding definitions.

Discrete Topology We also have what we call a discrete topology, which is the other side of spectrum. The discrete topology is the finest topology, since it says that all the subsets of the set of all open sets are be open.

$$\{\mathcal{O}_\alpha\} = \text{all subsets}$$

So clearly, all three criteria are already satisfied, because all the subsets are defined to be open. So trivially, this is already a topology. But the problem with this is that it's too fine grained, and it's not exactly useful anywhere else. That's a very rough sketch of what a topology is, let me tell you what topologies are used for.

A topology on a set is essential for providing a notion of continuity of maps on the set.

This is the most important point. We normally have this naive, sort of intuitive understanding of what a continuous function is, this is already distilled by mathematicians. So if you were to talk about an arbitrary set, what notion do you need for you to define the continuity of functions on a set.

So in other words, say we have a map f

$$f : M \to N$$

if you say for example that f is continuous map from M to N, the assumption here is that, you cannot make sense of this unless M and N are topological spaces. This is already assuming that M and N to be topological spaces. In other words, they need to be sets with a topology, i.e., both M and N have identifiable open sets that are subject to the conditions defined above.

This is all a bit a abstract, but hopefully will induce you to try to learn more because, topology, in general, is taught over several semesters.

So I'm just giving you a sense here of what it means, and because we need to define what continuity is, in order to grasp a full understanding of what we need topological spaces for.

3.1.4 Continuity

The modern definition of continuity is this

Definition 3.4 : Continuity

In general, a map $f : M \to N$ is continuous if the preimage of $\mathcal{O}$ belongs to the topology $\mathcal{T}_1$, for all open sets in $\mathcal{T}_2$, where $\mathcal{T}_1$ is the topology on M and $\mathcal{T}_2$ is the topology on N

$$f^{-1}[\mathcal{O}] \in \mathcal{T}_1, \ \forall \mathcal{O} \in \mathcal{T}_2$$

What does this all mean? If you say *the topology*, you don't mean just the set. Say the set M, it has the accompanying topology which is just a specification of open sets on M, then you have N, where you need to also specify what the open sets are (what you call $\mathcal{T}_2$). The requirement then, for the map f to be continuous is that for all the sets that you take in N, its preimage, must also be open in $\mathcal{T}_1$, or in M. This is the clearest way that it can be discussed which makes sure that you can understand that this is in fact identical to the notion of continuity in terms of open balls that is being discussed in calculus.

Suffice to say, then, that the definition of continuity requires topologies to exist in both the domain and the codomain.

$$\left.\begin{matrix}(M, \mathcal{T}_1) \\ (M, \mathcal{T}_2)\end{matrix}\right\} \text{Topological spaces}$$

The key point here is that without topology, there is no notion of continuity. Now this all sounds very abstract; I mean this is a fun subject, I actually like topology a lot, but for physics, it is usually enough to use open sets.

What is a preimage? The preimage is kind of like the inverse of a function. For example, say that f is a function which maps the domain M to the codomain N (Fig. 5). The preimage means that if you take any set in N, say $\mathcal{O}$, then the preimage of $\mathcal{O}$ is all the elements on M

that when f is applied, will be mapped to N. So, in essence, it's like the inverse function. The inverse of f need not exist, but the preimage will always exist. So there are a bunch of points in M, call them $\mathcal{W}$, that will be mapped entirely to $\mathcal{O}$ by f. Those sets of points is the preimage of $\mathcal{O}$. i.e., $f(f^{-1}[\mathcal{O}]) = \mathcal{O}$

$$f^{-1}[\mathcal{O}] = \{x \in M : f(x) \in \mathcal{O}\}$$

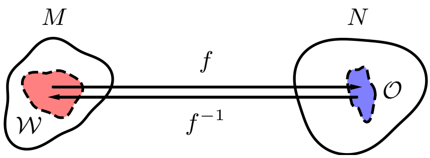


Figure 5: Pre-image of $\mathcal{O}$, $\mathcal{W} = f^{-1}[\mathcal{O}]$ and $\mathcal{O} = f[\mathcal{W}]$

Usual examples of Topologies We already discussed many weird examples like indiscrete and discrete topologies. However, there are more usual topologies. These are the so-called **Open ball topologies**, the standard topology in $\mathbb{R}^n$. Some examples are

(1) $\mathbb{R}$

The real line is an example of an open topology. It has open intervals as its open sets; this is the standard Euclidean topology. Such examples are the set of numbers from x to y in the real line:

$$(x, y), \quad x < y, \quad x, y \in \mathbb{R}$$

These are the open sets. Therefore, the set of open sets of $\mathbb{R}$ is

$$\{\mathcal{O}_\alpha\} = \{\cup_\alpha (x_\alpha, y_\alpha) : x_\alpha < y_\alpha\}$$

These are the open intervals which define a topology. You can check this using the three criteria: $\mathbb{R}$ is open $(-\infty, \infty)$, any union of open intervals is still open, and any finite intersections of open intervals in the real line is also open. Therefore, these sets form a topology on $\mathbb{R}$.

Remark :

Finite here is important, because you can imagine a construction of sets, say $\left\{\frac{1}{n}, 1\right\}$, $n \in \mathbb{N}$, these are all individually open, this is a subset of all possible intervals from 0 to 1. If you take an infinite intersection of this interval, $1/n$ will eventually converge (limit) to zero, hence:

$$\bigcap_{n}^{\infty}\left(\frac{1}{n}, 1\right) = [0, 1)$$

Then it is no longer open. For larger and larger n's, the result will include 0, so it will longer be open, that's why you need the intersections to be finite. That's the standard topology in $\mathbb{R}$. In fact, you can also define a topology using closed intervals, that's also an interesting exercise.

(2) $\mathbb{R}^n$

Typically, in general $\mathbb{R}^n$, we use open balls B_r with radius r.

$$B_r(\mathbf{x}) \equiv \{\mathbf{y} : |\mathbf{x} - \mathbf{y}| < r\}$$

That is, all the points $\mathbf{y}$ with a distance from $\mathbf{x}$ less than r, a *ball* which has open intervals, is an open set.

If you declare that your open sets are open balls in $\mathbb{R}^n$, you recover the so-called standard topology as well. Similar to the arguments for $\mathbb{R}^1$, we can also show that all of $\mathbb{R}^n$, $n \in \mathbb{Z}$ to be topologies. It has a name, called Euclidean topologies.

There's a lot to study here, but we're not going to need all of topology, it's sufficient to know what a topology is and why it's important for our understanding of a manifold.

So again, what is a topology? It's just a set and a specification of open sets of the set. That is, $\mathcal{M}$ and the open sets of $\mathcal{M}$.

Going back to the definition of a manifold, it is a topological space with an open cover $\{\mathcal{O}_\alpha\}$, these are just the open sets I talked about, and also, we require that the union of all these open sets, it results to the full manifold set itself, $\mathcal{M}$.

$$\mathcal{M} = \bigcup_{\alpha} \mathcal{O}_\alpha \tag{3.1.2}$$

There should be no element of the set that does not belong to an open set. For example, you start with a set $\mathcal{M}$, and then you declare open sets to be your topology. In that declaration, there should be no element or point in $\mathcal{M}$ that is not part of an open set. That's the requirement for you to have an open cover. These sets of open sets need to satisfy the conditions for the criteria for a topological space.

It has charts, which maps open sets to a subset of $\mathbb{R}^n$, and it is a homeomorphism. Homeomorphism just means that f and f^{-1} are both continuous. It is a rough sense of saying that they are topologically equivalent spaces. So it's saying that those open sets are topologically the same as $\mathbb{R}^n$.

3.1.5 Homeomorphism

Let's define a homeomorphism; Now that we have a notion of continuity, we can use that.

Definition 3.5 : Homeomorphism

A map f between two sets M to N, is said to be *homeomorphic* or a homeomorphism, if

$$f : M \to N$$

such that f and its preimage f^{-1} are both continuous.

So a homeomorphism is usually what we understand is the precise terminology for a map that continuously deforms something to something else; That's how we usually understand topology, it's like rubber sheet geometry. We demand a unique, one-to-one correspondence from points from M to N. If you can continuously deform something to something else[15], then those two things are homeomorphic, they are deformable to each other. This is best illustrated by Fig. 6.

We take this to mean that M and N are topologically equivalent. In the point of view of topology, they are one and the same, you cannot distinguish them with topological features, i.e., they are continuously deformable to each other.

[15] No cutting or removing parts, all parts of N can be remapped to M

Figure 6: A classic example of a homeomorphism: A topology joke

3.1.6 Charts

A requirement for manifolds is that for all open sets in the manifolds, there is an attached chart. This chart has to be a homeomorphism from the open sets that you've declared on your set, to a subset of $\mathbb{R}^n$, which would be some union of open balls because open balls will have a standard topology.

Suppose you have open sets $\mathcal{O}_1, \mathcal{O}_2$, and $\mathcal{O}_3$ on a manifold $\mathcal{M}$. If you take the union of all these things that you've declared to be open, then it should become the entire manifold, from (3.1.2). Then it constitutes an open cover. For each of these sets $\mathcal{O}_\alpha$, there should be a map ψ_α from $\mathcal{O}_\alpha$ to $\mathbb{R}^n$. We can only represent these as two dimensional but these are n dimensional. This is illustrated by Fig. 7

For example, take $\mathcal{O}_\alpha$, it has a chart ψ_α that maps it to $\mathbb{R}^n$. It therefore has an image $\psi_\alpha[\mathcal{O}_\alpha]$ under ψ_α

$$\psi_\alpha : \mathcal{O}_\alpha \to \psi_\alpha[\mathcal{O}_\alpha] \subseteq \mathbb{R}^n$$

Now, this may sound abstract but this is something that we do in physics all the time. Because this process of providing this ψ_1 which is a map from points from $\mathcal{M}$ to $\mathbb{R}^n$, that's nothing but a coordinate system. $\mathcal{M}$ is just a set, an abstract set with no coordinates, it's an independent object on its own, simply a collection of points constituting the elements

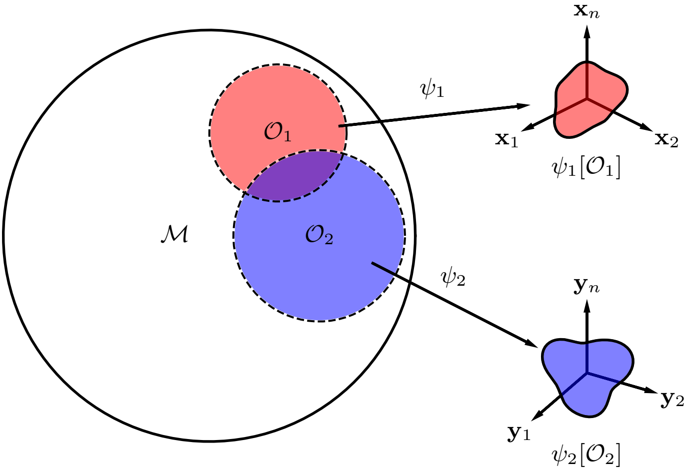


Figure 7: Charts ψ mapping from $\mathcal{O}_\alpha$ to $\mathbb{R}^n$

of $\mathcal{M}$. There is no innate coordinates there. When you supply it with coordinates, as we do in physics all the time, that's where the notion of manifolds comes in.

Manifolds are used here because we're giving each point in an abstract set $\mathcal{M}$, say p^i, a set of coordinates $\mathbf{x}_i$. In other words we're giving a map from $\mathcal{O}_1$ to have some numerical value of each point, similar to how we assign points on the Earth (an abstract oblate-spheroidal collection of atoms and points) arbitrary longitudes and latitudes ($\mathbb{R}^2$). That is, we are *charting* the manifold.

$$\psi : \mathcal{O} \to \mathbb{R}^n, \quad p \mapsto \mathbf{x} = (x^1(p), \ldots, x^n(p))$$

Now, the requirement that ψ_1 needs to be a homeomorphism is a statement that $\mathcal{O}_1$ and $\psi_1[\mathcal{O}_1]$ are topologically the same. What you did is you continuously deformed $\mathcal{O}_1$ into $\psi_1[\mathcal{O}_1]$, but they are topologically equivalent. There is no difference.[16] That's all basically what it is.

These charts are otherwise known in physics as coordinate systems. This is what I call locally $\mathbb{R}^n$. The coordinates that ψ_1 uses, only apply to

[16] Jesus fucking christ all that work just to say "we just put coordinates in objects"

$\mathcal{O}_1$. If we take a point outside $\mathcal{O}_1$, say from $\mathcal{O}_2$, then ψ_1 is no longer valid. It's only locally $\mathbb{R}^n$, because $\mathcal{O}_2$ will have a different $\mathbb{R}^n$ appropriate to it. So that the coordinates that ψ_1 furnish on $\mathcal{O}_1$ will no longer apply to any point that does not belong to $\mathcal{O}_1$. The map is no longer valid.

The final condition for a manifold to be space time is that it has to be differentiable, i.e., how the different coordinate systems interact with each other.

3.1.7 Differentiable Manifolds

Differentiable or smooth manifolds (both are synonymous) i.e., C^∞ manifolds, are manifolds which can be differentiated continuously. What do we mean by that?

Take an open set $\mathcal{O}_1$ and $\mathcal{O}_2$ both subsets of $\mathcal{M}$, and overlap each other, i.e., there is a part of $\mathcal{O}_2$ that is a subset of $\mathcal{O}_1$ and vice versa. ψ_1 maps $\mathcal{O}_1$ to $\mathbb{R}^n$, say $\mathbf{x} = (x_1, x_2, \ldots, x_n)$, and ψ_2 maps $\mathcal{O}_2$ to $\mathbb{R}^n$, say $\mathbf{y} = (y_1, y_2, \ldots, y_n)$. Note that these are different coordinate systems that each apply only to the chart which mapped it. However, since $\mathcal{O}_1$ and $\mathcal{O}_2$ are intersectional, then the maps must also have some parts that are intersectional. That is, ψ_1 will map both parts of $\mathcal{O}_1$ and $\mathcal{O}_2$. As such, there will be a map that is $\psi_1[\mathcal{O}_1 \cap \mathcal{O}_2]$. Similarly, ψ_2 will map a part of $\mathcal{O}_1$ intersectional to $\mathcal{O}_2$, $\psi_2[\mathcal{O}_2 \cap \mathcal{O}_1]$.

Notice that you can actually map both these regions from $\psi_2[\mathcal{O}_2 \cap \mathcal{O}_1]$ to $\psi_1[\mathcal{O}_1 \cap \mathcal{O}_2]$ directly. You can do that by constructing:

$$\psi_1 \circ \psi_2^{-1} : \mathbb{R}^n \to \mathbb{R}^n \tag{3.1.3}$$

Obviously ψ_2^{-1} only works on the region that was mapped by ψ_2. We can then map the preimage of $\psi_2[\mathcal{O}_1 \cap \mathcal{O}_2]$, which is defined, then composing that to the map ψ_1 to $\mathbb{R}^n$, i.e., it maps from $\mathbb{R}^n$ to another $\mathbb{R}^n$, thereby bypassing the need for explicitly describing $\mathcal{M}$.

Equation (3.1.3) only really applies to the region $\mathcal{O}_1 \cap \mathcal{O}_2$. As such, the map $\psi_1 \circ \psi_2^{-1}[\mathcal{O}_1 \cap \mathcal{O}_2]$ only really applies to $\psi_2[\mathcal{O}_1 \cap \mathcal{O}_2]$, not to all of $\mathbb{R}^n$.

$$\psi_1 \circ \psi_2^{-1} : \underbrace{\psi_2[\mathcal{O}_1 \cap \mathcal{O}_2]}_{\subseteq \mathbb{R}^n} \to \underbrace{\psi_1[\mathcal{O}_1 \cap \mathcal{O}_2]}_{\subseteq \mathbb{R}^n}$$

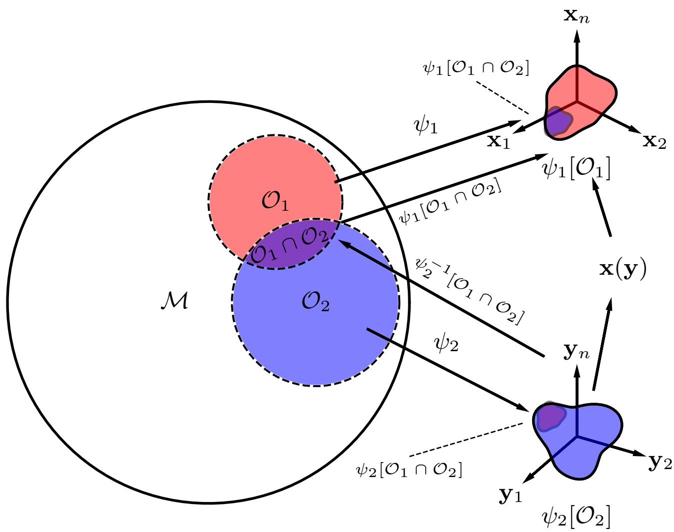


Figure 8: Differentiability of manifolds

This can be written in the more concrete usual notation as $\mathbf{x}$ being a function of $\mathbf{y}$, i.e., taking the $\mathbf{y}$ coordinates and spitting out the $\mathbf{x}$ coordinates:

$$\psi_1 \circ \psi_2^{-1} \iff \mathbf{x}(\mathbf{y}) \tag{3.1.4}$$

These are called **transition functions** in the math literature. But for now, we'll call it, simply, as coordinate transformations.

Now, let's go back to the definition of a manifold, Def. 3.3, we see that *all such transition functions* $\psi_\beta \circ \psi_\alpha^{-1}$ *must be* C^∞.

This is the last key part, and we have now explained all the terms needed to define a differentiable manifold (a.k.a., C^∞ manifolds). The definition of a C^∞ manifold requires that all such coordinate transformations are C^∞, and we understand C^∞ in the usual sense, because in normal calculus, you can take as many derivatives as you want, and still be continuous. This is applicable because $\mathbf{x}(\mathbf{y})$ is a usual $\mathbb{R}^n \to \mathbb{R}^n$ function, it is something that everybody understands. All objects like these should be infinitely differentiable. One should take as many derivatives as they want.

So again, what is a manifold? It's simply a set of open sets with coordinate systems, and every time the open sets overlap, the transition functions (coordinate transformations) must always be infinitely differentiable, if it is to be a C^∞ manifold.

Question: About the notion of C^∞, if you take derivatives again and again of a function, wouldn't it at some point become zero? And so, it seems impossible for it to be infinitely differentiable.

It is still differentiable! If you can differentiate a zero function, it is still differentiable, and the result is 0 again, *ad infinitum*, you can still keep on going, and the result remains continuous. What we mean by differentiability here is the usual criteria for differentiability which are the lack of spikes or discontinuities that will make the manifold undifferentiable at that point. We require these to be *smooth*. When we get to black holes, we will find that the coordinate systems will break down precisely when you encounter violations of this requirement to be infinitely differentiable, which signifies we reached the end of our chart, and we need a new coordinate system.

One of the things which you should sort of take from this notion of a manifold is that the coordinate systems can lead you astray. The manifolds can be very well defined and well behaved. However, at some point, the transformation will cause a divergence which is not allowed by your chosen coordinate transformation, that will signify you have a problem with the chart, it has gone beyond where it can represent, by which you need to find another coordinate transformation which fits your uses. There are other kinds of manifolds which are not C^∞, mathematicians may sometimes demand only C^2. So sometimes you may see people say C^2 manifold, because for the transition functions, they only want it to be C^2. Sometimes they say topological manifold, this is simply a C^0 manifold, it is only continuous but the derivatives are not. For us in physics, we're always going to use infinitely differentiable manifolds. If we see a manifold, we will always demand for it to be a C^∞, or a smooth, or differentiable manifold. This is what we demand of spacetime.

The entire set of open-set/chart pairs, since it is an open cover, is called an atlas (to continue the analogy with maps)

$$\text{Atlas} \equiv \{(\psi_\alpha, \mathcal{O}_\alpha)\}$$

The unions of $\mathcal{O}_\alpha$ and ψ_α constitute an atlas on S^2

3.1.8 Examples

(1) $\mathbb{R}^n$

Obviously. $\mathbb{R}^n$ itself is a manifold. In fact, the nice thing about $\mathbb{R}^n$ is that it can be covered by a single chart. For $\mathbb{R}^n$, it's just the usual $\mathbb{R}^n$. This chart is that the identity transformation:

$$\text{Id} : \mathbb{R}^n \to \mathbb{R}^n$$

(2) S^2

S^2, the two sphere, has various ways to coordinatize. One way is by the so-called stereographic projection (Fig. 9). We want a chart ψ that globally maps all of S^2 to R^2, such that $\psi : S^2 \to \mathbb{R}^2$. One nice map is called the stereographic projection, where we coordinatize points by using the north pole N, and projecting a line from N to any point in S^2, p until it intersects a plane described by $\mathbb{R}^2$. We can place the plane to be coincident with the center of S^2 or on the south pole S. The intersection point on the plane, (x, y) is then the map from the surface of S^2 to $\mathbb{R}^2$.

$$\psi_N : p \mapsto (x, y)$$

This is simply a way of assigning S^2 to $\mathbb{R}^2$. However, since we map S^2 from the point N, it cannot be projected onto the plane, hence, it is not represented in $\mathbb{R}^2$

$$\psi_N : S^2 - \{N\} \to \mathbb{R}^2$$

This is an example of a coordinate system which can have multiple charts. ψ_N is insufficient to describe all of S^2, hence, we need another map. We will construct the same projection, where we attach the plane to be coincident with N, and instead take the line from S to project to the plane. This is now

$$\psi_S = S^2 - \{S\} \to \mathbb{R}^n$$

We call $S^2 - \{S\} = \mathcal{O}_S$ and $S^2 - \{N\} = \mathcal{O}_N$. Note that the union is the entire manifold.

$$\mathcal{O}_N \cup \mathcal{O}_S = S^2$$

So, $\{\psi_N, \mathcal{O}_N\}$ and $\{\psi_S, \mathcal{O}_S\}$ constitute an atlas for S^2.

Another way is to use longitude and colatitude, the usual way to coordinatize a sphere

$$(\theta, \phi)$$

where θ is the colatitude, $\theta = [0, \pi]$, and ϕ is the longitude, ($\phi = [0, 2\pi)$). With this method, we can also embed S^2 to $\mathbb{R}^3$. This chart maps all of S^2 to a *subset* of $\mathbb{R}^2$. However, it has two main problems: (1) At the poles, longitude is undefined, for every longitude line meets there, therefore ϕ is undefined, and the system breaks down, and (2) The longitude wraps around, with 0 and 2π representing the same meridian, so the mapping repeats.

However, it is a very useful coordinate system on most of the sphere, but not a global chart. In that sense, other maps are needed to complete the entire atlas. This illustrates why a manifold may require multiple coordinate charts to describe it globally. For most intents and purposes, using the latitude-longitude convention is very useful. However, due to it's limitations in describing all points on S^2, other projection mappings are preferred.

These are just two ways to map S^2, there are other ways to do so which are more contrived, but I encourage you to study them

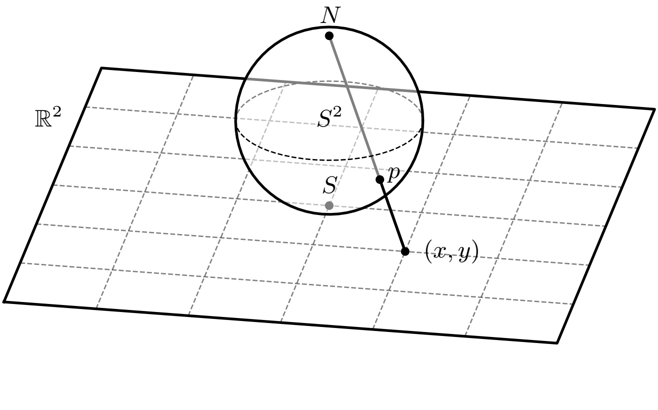


Figure 9: Stereographic projection of S^2 to $\mathbb{R}^2$, using N

Like we said in the beginning, literally anything in physics where we use coordinates is a manifuld. Normally in most areas of physics, we're not too careful about the manifold because we mostly think about the stuff on top of the manifold. However, this is spacetime physics, when we study spacetime, spacetime itself *is* the manifold. So we have to consider the manifold itself. Anything you might conceive of, field theory (fields living in $\mathbb{R}^4$, coordinatized by t and $\mathbf{x}$), mechanics (which lives in phase space, coordinatized by x and p), thermodynamics, which are also manifolds coordinatized by thermodynamic variables. Literally anything where you're using coordinates, there is an underlying manifold there.

Very roughly, we wouldn't be too far off from the definition that basically a manifold is something that it's a set and that it's locally coordinatizable, and that locally is equivalent to $\mathbb{R}^n$. If you can put coordinates on a set, you essentially have a manifold.

Now, obviously the question would be: What are not manifolds? They do exist, those are the sets which does not have the things we demand of them. For example, A geometric object that appears to self-intersect in an embedding may fail to be a manifold at the intersection point, for the definitiones required will fail at the interseciton.

In physics, GR especially, we respect the democracy of coordinate systems. Charts are just there as computational devices. They have to interact with charts in such a way that we get coodinate transformations, infinitely continuous. If your transformation is not continuously differentiable, which means you have a divergence somewhere, it means that your chart is invalid. It signals that the domain of your chart is probably smaller than you imagined.

3.2 Structures on Manifolds

We begin this meeting by recapping what we learned from the previous lecture, namely differentiable manifolds, charts, and topology. We then develop the notion of spacetime as a differentiable manifold and discuss the importance of describing the laws of physics in terms of geometric objects rather than particular coordinate systems. From there, we introduce scalar fields and, more importantly, develop the definition of vectors on a general manifold through the idea of directional derivatives.

3.2.1 Spacetime as a Manifold

We consider spacetime to be a differentiable, or smooth, manifold. That is, spacetime is a topological space which is *locally* homeomorphic to $\mathbb{R}^n$, together with additional structure that allows us to perform calculus.

The important point is that a manifold does not, in general, possess a single global coordinate system. Instead, we describe it using coordinate patches, or *charts*, each of which maps a portion of the manifold into $\mathbb{R}^n$.

For example, consider a three-dimensional manifold $\mathcal{M}$. Suppose that $Q_1, Q_2 \subset \mathcal{M}$ are two overlapping coordinate patches. On Q_1, we may introduce spherical coordinates

$$\mathbf{r} = (r, \theta, \phi),$$

while on Q_2 we may introduce Cartesian coordinates

$$\mathbf{x} = (x, y, z).$$

A chart is a pair

$$\{Q_\alpha, \psi_\alpha\},$$

where ψ_α is a map from the manifold to $\mathbb{R}^n$. It is a coodinate system, so to speak.

$$\psi_\alpha : Q_\alpha \to \mathbb{R}^n$$

maps points on the manifold to their coordinate representations.

Thus, if $p \in Q_1 \cap Q_2$, we can describe the same point using either coordinate system:

$$\psi_1(p) = \mathbf{r}, \qquad \psi_2(p) = \mathbf{x}.$$

Since the two charts overlap, we can move from one coordinate representation to the other by composing one chart with the inverse of the other:

$$\mathbf{x} = \psi_2 \circ \psi_1^{-1}(\mathbf{r}). \tag{3.2.1}$$

Likewise,

$$\mathbf{r} = \psi_1 \circ \psi_2^{-1}(\mathbf{x}). \tag{3.2.2}$$

More directly, we can represent $\mathbf{x}$ as a function of $\mathbf{r}$, and vice versa.

$$\mathbf{x}(\mathbf{r}) \quad \longleftrightarrow \quad \mathbf{r}(\mathbf{x})$$

The maps

$$\psi_\beta \circ \psi_\alpha^{-1}$$

are called *transition maps.* The requirement that these transition maps be sufficiently differentiable is what allows the manifold to be given a differentiable structure.

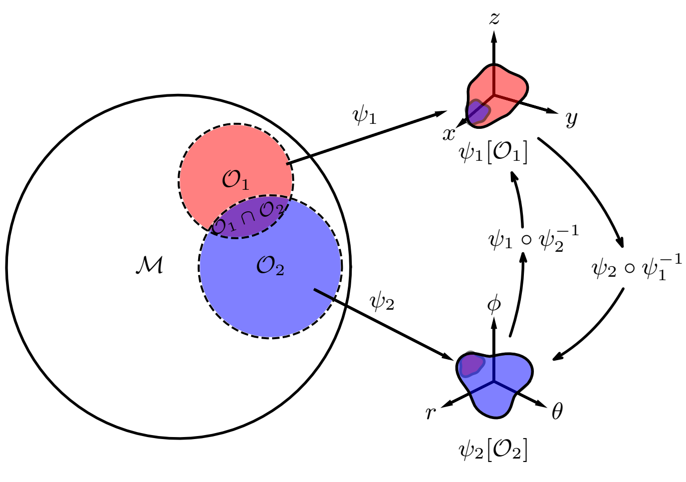


Figure 10: An example of a manifold being represented in different coordinate systems, cartesian and polar in this case, and how to interchange between them using transition functions.

A collection of charts whose domains cover the manifold,

$$\mathcal{M} = \bigcup_{\alpha} Q_\alpha,$$

is called an *atlas*. The charts are required to be compatible on their overlaps:

$$\psi_\beta \circ \psi_\alpha^{-1} : \psi_\alpha(Q_\alpha \cap Q_\beta) \to \psi_\beta(Q_\alpha \cap Q_\beta) \tag{3.2.3}$$

must be smooth wherever the two coordinate patches overlap.

This is the essential idea behind a differentiable manifold. We do not require the manifold itself to resemble $\mathbb{R}^n$ globally. We only require that, in sufficiently small neighborhoods, it can be described using ordinary Euclidean coordinates.

The existence of multiple charts is particularly useful when a coordinate system develops a *coordinate singularity*. If one coordinate system becomes inconvenient or singular in some region, we may simply use another chart which covers that region and the singularity does not exist.

For example, spherical coordinates are singular at $r = 0$ and at the poles where $\theta = 0, \pi$. These singularities need not represent anything physically pathological; they may simply indicate that the chosen coordinates have broken down.

In general relativity, this distinction is crucial. A coordinate singularity is not necessarily a physical singularity, it may only be a mathematical artifact. For example, the Schwarzschild coordinates possess a coordinate singularity at $r = 2GM$ (in units where $c = 1$), which can be removed by changing coordinates. By contrast, the Schwarzschild curvature singularity at $r = 0$ is a genuine physical singularity and is not removed by a mere change of coordinates.

3.2.2 General Covariance

The central lesson of the manifold formulation is that the laws of physics should not depend on the particular coordinate system used to describe them. As long as the coordinate system is appropriate, one may use any system he or she desires that makes the physics being performed more conveniet, as long as the limits of such formulation is known.

In general relativity, physical quantities should therefore be expressed in terms of geometric objects which possess well-defined transformation properties under changes of coordinates.

Among the objects we will encounter are:

(a) scalars,

(b) vectors,

(c) one-forms, or dual vectors,

(d) tensors,

(e) spinors.

Spinors require some additional structure beyond the ordinary tensor bundle and therefore do not fit directly into the elementary tensor description of general relativity. Nevertheless, spinor fields can be defined on suitable spacetime manifolds and are important when describing fermionic matter.

The purpose of this meeting is to construct these geometric objects starting from the manifold itself.

3.2.3 Scalar Functions

A scalar field is the simplest geometric object that we can define on a manifold. It is simply a function which assigns one real number to every point of the manifold:

$$f : \mathcal{M} \to \mathbb{R}. \tag{3.2.4}$$

There is no coordinate system involved in the definition itself.

Given a point p in the manifold.

$$p \in \mathcal{M},$$

the scalar field simply assigns p to a real number

$$f : p \mapsto f(p) \in \mathbb{R}.$$

Coordinates only enter when we wish to represent the scalar field explicitly. Given a chart

$$\psi : Q \subset \mathcal{M} \to \mathbb{R}^n,$$

we can construct the coordinate representation of f by composing it with the inverse chart:

$$f \circ \psi^{-1} : \mathbb{R}^n \to \mathbb{R}. \tag{3.2.5}$$

Writing

$$\mathbf{x} = \psi(p) = \left(x^1, x^2, \dots, x^n\right),$$

we therefore obtain the familiar expression of a scalar field.

$$f : \mathbf{x} \mapsto f \circ \psi^{-1}(\mathbf{x}) \equiv f(\mathbf{x}). \tag{3.2.6}$$

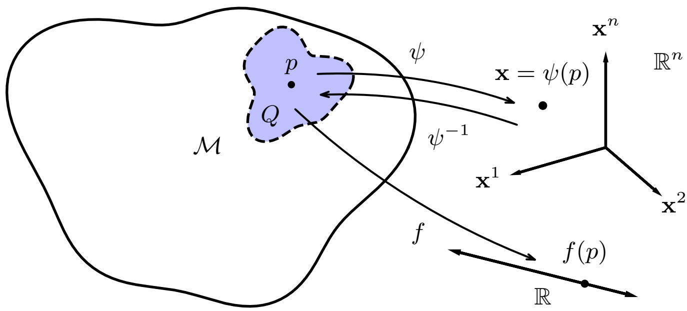


Figure 11: A scalar function as a map from a manifold to the real numbers.

The coordinate representation of the function arises only after choosing a chart. Thus, the coordinates

$$\mathbf{x} \equiv (x^1, x^2, \dots, x^n)$$

should not be confused with points of the manifold themselves. They are merely the coordinate representation of points in a particular chart. The manifold itself is coordinate-independent.

To perform differential geometry, we require our functions to possess the appropriate degree of differentiability. A function is called *smooth* if its coordinate representation is infinitely differentiable:

$$f \in C^\infty(\mathcal{M}). \tag{3.2.7}$$

We may denote the set of all smooth real-valued functions on $\mathcal{M}$ by

$$\mathcal{F}(\mathcal{M}) \equiv C^{\infty}(\mathcal{M}). \tag{3.2.8}$$

An important example of scalar functions are the coordinate functions themselves. Given a chart ψ, we may write

$$x^{\alpha}(p) \equiv \psi^{\alpha}(p), \tag{3.2.9}$$

where ψ^{α} denotes the αth component of the chart.

Thus, each coordinate x^{α} is itself a scalar function on the region covered by the chart. This point will become important when we construct vectors.

3.2.4 Vectors

The concept of a vector space is undoubtedly familiar to most readers. In prerelativity physics it is assumed that space has the natural structure of a three dimensional vector space once one has designated a point to serve as the origin; the natural rules for adding and scalar multiplying spatial displacements satisfy the vector space axioms. In special relativity, spacetime similarly has the natural structure of a four-dimensional vector space.

However, when one considers curved geometries (as we do in general relativity), this vector space structure is lost. For example, there is no natural notion of how to *add* two points on a sphere and end up with a third point on the sphere. Nevertheless, vector space structure can be recovered in the limit of "infinitesimal displacements" about a point. It is this notion of "infinitesimal displacements" or tangent vectors which lies at the foundation of calculus on manifolds. Therefore, we will devote considerable attention below to giving a precise mathematical definition of this concept.

The ordinary notion of a vector comes from linear algebra.

A vector is an algebraic object that belongs to a vector space $\mathbb{V}$, which is an algebraic structure equipped with operations of vector addition and scalar multiplication satisfying the usual vector-space axioms, which defines what multiplication and addition is, and what is a unit and a zero vector, among other things. However, in this case, we are concerned about closure and linearity.

In particular, if

$$\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{V}, \qquad \alpha, \beta \in \mathbb{R},$$

then closure requires

$$\alpha\mathbf{v}_1 + \beta\mathbf{v}_2 \in \mathbb{V}. \tag{3.2.10}$$

Anything can be called a vector provided that it satisfies all the axioms defining a vector space. There is therefore no requirement that a vector must necessarily be represented as an arrow in Euclidean space.

The familiar arrow picture is a useful visualization, but it is not the fundamental definition of a vector.

In $\mathbb{R}^3$, we normally introduce vectors as directed line segments. Given points p and q, for example, one can draw an arrow pointing from p to q.

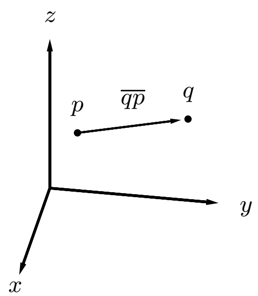


Figure 12: The familiar representation of vectors as directed arrows in Euclidean space. This picture relies on the affine structure of $\mathbb{R}^3$.

There is, however, a subtlety here.

In Euclidean space, we possess an *affine structure*: points can be subtracted to obtain vectors, and vectors can be added to points. Thus, if p and q are points in $\mathbb{R}^3$, the displacement

$$q - p \equiv \overline{qp}$$

is naturally interpreted as a vector.

A general manifold does not possess this structure. It is not applicable or intrinsic to the manifold itself. In particular, there is no natural operation $q - p$ for two arbitrary points $p, q \in \mathcal{M}$.

Therefore, if we want to define vectors on a general manifold, we need a definition which does not depend upon arrows, position vectors, or the ability to subtract points.

The solution is to define vectors in terms of *directional derivatives.*

3.2.5 Directional Derivatives

In $\mathbb{R}^n$, suppose that $\mathbf{v}_p$ is a vector based at the point p. Given a smooth function f, its directional derivative in the direction $\mathbf{v}_p$ is

$$\mathbf{v}_p(f) \equiv \lim_{t\to 0} \frac{f(\mathbf{x}_p + t\mathbf{v}_p) - f(\mathbf{x}_p)}{t}. \tag{3.2.11}$$

In ordinary vector calculus, this is equivalent to

$$\mathbf{v}_p(f) = \mathbf{v}_p \cdot \nabla f(p) \tag{3.2.12}$$

However, Eq. (3.2.11) makes use of the affine structure of $\mathbb{R}^n$, since it explicitly contains

$$\mathbf{x}_p + t\mathbf{v}_p.$$

We therefore seek a definition which retains the algebraic properties of a directional derivative without requiring us to add vectors to points.

The key observation is that the directional derivative is an *operator acting on functions.*

Consider a vector $\mathbf{v}_p$ acting on two smooth functions f and g. Directional differentiation satisfies two fundamental properties.

(1) It is linear:

$$\mathbf{v}_p(\alpha f + \beta g) = \alpha \mathbf{v}_p(f) + \beta \mathbf{v}_p(g), \tag{3.2.13}$$

for

$$\alpha, \beta \in \mathbb{R}, \qquad f, g \in \mathcal{F}(\mathbb{R}^n)$$

(2) It satisfies the Leibniz rule:

$$\mathbf{v}_p(fg) = f(p)\mathbf{v}_p(g) + g(p)\mathbf{v}_p(f). \tag{3.2.14}$$

These two properties do not require an affine structure.

We therefore take them as the fundamental definition of a tangent vector.

Definition 3.6 : Tangent Vector

A tangent vector at $p \in \mathcal{M}$ is a linear map of manifolds to real functions

$$\mathbf{v}_p : \mathcal{F}(\mathcal{M}) \to \mathbb{R}^n \tag{3.2.15}$$

which satisfies the Leibniz rule

$$\mathbf{v}_p(fg) = f(p)\mathbf{v}_p(g) + g(p)\mathbf{v}_p(f). \tag{3.2.16}$$

Thus, we define vector is nothing but a derivative operator on a manifold satisfying the appropriate algebraic properties.

This also explains why the usual arrow does not appear in the definition. The arrow is only our preconceived geometric picture of a vector. The fundamental object is the action of the vector on functions.

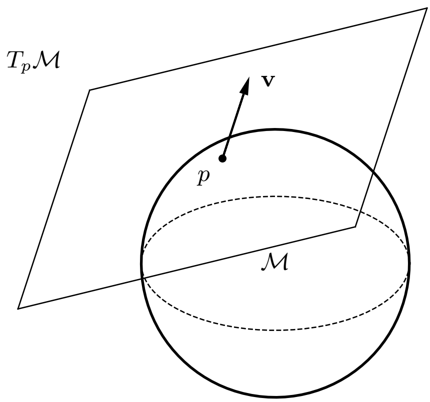


Figure 13: A tangent vector $\mathbf{v}$, represented as a directional derivative lying in the tangent plane at a point p of the manifold, in this case, a surface embedded on a sphere in $\mathbb{R}^3$. The tangent plane in this case represents the tangent space $T_p\mathcal{M}$

.

3.2.6 The Tangent Space

The collection of all tangent vectors at a point p forms a vector space.

Theorem 3.1 : Tangent Space

The set of all tangent vectors at $p \in \mathcal{M}$ forms a vector space, called the *tangent space at p*, denoted

$$T_p\mathcal{M}. \tag{3.2.17}$$

Proof. Let $\mathbf{v}_p, \mathbf{w}_p \in T_p\mathcal{M}$.

We define vector addition by the plus operator, $+$. It acts likeso

$$(\mathbf{v}_p + \mathbf{w}_p)(f) \equiv \mathbf{v}_p(f) + \mathbf{w}_p(f), \tag{3.2.18}$$

and scalar multiplication by the following operation:

$$(\alpha\mathbf{v}_p)(f) \equiv \alpha\mathbf{v}_p(f). \tag{3.2.19}$$

The zero vector is defined by

$$\mathbf{0}_p(f) = 0. \tag{3.2.20}$$

We must verify closure. For the sum,

$$\begin{aligned}(\mathbf{v}_p + \mathbf{w}_p)(fg) &= \mathbf{v}_p(fg) + \mathbf{w}_p(fg) \\ &= f(p)\mathbf{v}_p(g) + g(p)\mathbf{v}_p(f) \\ &\quad + f(p)\mathbf{w}_p(g) + g(p)\mathbf{w}_p(f) \\ &= f(p)(\mathbf{v}_p + \mathbf{w}_p)(g) + g(p)(\mathbf{v}_p + \mathbf{w}_p)(f). \end{aligned} \tag{3.2.21}$$

Hence $\mathbf{v}_p + \mathbf{w}_p$ also satisfies the Leibniz rule and is therefore a tangent vector.

Similarly, for $\alpha \in \mathbb{R}$,

$$\begin{aligned}(\alpha\mathbf{v}_p)(fg) &= \alpha\mathbf{v}_p(fg) \\ &= \alpha f(p)\mathbf{v}_p(g) + \alpha g(p)\mathbf{v}_p(f) \\ &= f(p)(\alpha\mathbf{v}_p)(g) + g(p)(\alpha\mathbf{v}_p)(f). \end{aligned} \tag{3.2.22}$$

Therefore the set of tangent vectors satisfies closure under addition and scalar multiplication, and the remaining vector space axioms follow from the corresponding properties of the real numbers. □

From these, we also prove that the corollary expressions are also linear.

$$(\alpha\mathbf{v}_p + \beta\mathbf{w}_p)(f) = \alpha\mathbf{v}_p(f) + \beta\mathbf{w}_p(f)$$

and

$$(\alpha\mathbf{v}_p + \beta\mathbf{w}_p) \cdot \nabla f = \alpha\mathbf{v}_p \cdot \nabla f + \beta\mathbf{w}_p \cdot \nabla f$$

Every vector in $T_p\mathcal{M}$ is therefore associated with the same point p. This is different from the elementary picture of vectors in $\mathbb{R}^n$, where vectors are often drawn with their tails at the origin.

A tangent vector does not have to be thought of as an arrow extending from one point of the manifold to another. It belongs to the tangent space *at a particular point.*

3.2.7 The Coordinate Basis

We now construct an explicit basis for $T_p\mathcal{M}$.

Suppose that

$$\{Q, \psi\}$$

is a chart around p, with coordinates

$$\{x^1, x^2, \ldots, x^n\}.$$

We define the coordinate basis vectors by their action on a smooth function f:

$$(\partial_\mu)_p(f) \equiv \left.\frac{\partial(f \circ \psi^{-1})}{\partial x^\mu}\right|_{\psi(p)}. \tag{3.2.23}$$

The notation

$$(\partial_\mu)_p$$

should therefore be interpreted as a map acting on functions.

In other words,

$$(\partial_\mu)_p : \mathcal{F}(\mathcal{M}) \to \mathbb{R}.$$

The derivative is taken not directly on the abstract manifold, but on the coordinate representation $f \circ \psi^{-1}$ in $\mathbb{R}^n$.

The remarkable fact is that these coordinate derivative operators form a basis for the entire tangent space.

Theorem 3.2 : Coordinate Basis of the Tangent Space

Given a chart $\{Q, \psi\}$ with coordinates $\{x^1, \ldots, x^n\}$, the set

$$\{(\partial_1)_p, (\partial_2)_p, \ldots, (\partial_n)_p\} \tag{3.2.24}$$

constitutes a basis for $T_p\mathcal{M}$.

Proof. We prove linear independence and spanning separately.

(1) Linear Independence

Recall that vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ are linearly independent if

$$\alpha^1\mathbf{v}_1 + \alpha^2\mathbf{v}_2 + \cdots + \alpha^n\mathbf{v}_n = \mathbf{0} \qquad \equiv a^\mu(\partial_\mu) = \mathbf{0}$$

implies that

$$\alpha^1 = \alpha^2 = \cdots = \alpha^n = 0 \qquad \equiv a^\mu = 0$$

In this case, we shall prove that is indeed the case.

Suppose that some coefficients α^μ and the basis sums to zero.

$$\alpha^\mu(\partial_\mu)_p = \mathbf{0}_p \tag{3.2.25}$$

Acting on an arbitrary function f gives

$$\alpha^\mu(\partial_\mu)_p(f) = 0. \tag{3.2.26}$$

In particular, since it is convenient, we choose $f = x^\nu$. Then

$$\alpha^\mu(\partial_\mu)_p(x^\nu) = 0. \tag{3.2.27}$$

By definition,

$$\begin{aligned}(\partial_\mu)_p(x^\nu) &= \left.\frac{\partial(x^\nu \circ \psi^{-1})}{\partial x^\mu}\right|_{\psi(p)} \\ &= \left.\frac{\partial x^\nu}{\partial x^\mu}\right|_p \\ &= \delta^\nu{}_\mu\end{aligned} \tag{3.2.28}$$

Hence,

$$\alpha^\mu \delta^\nu{}_\mu = \alpha^\nu \\ = 0 \tag{3.2.29}$$

Therefore the coordinate basis vectors are linearly independent.

(2) Spanning[17]

Let

$$\mathbf{v}_p \in T_p\mathcal{M}.$$

Define the components of $\mathbf{v}_p$ with respect to the coordinate basis as the action of the operator on that basis.

$$v^\nu \equiv \mathbf{v}_p(x^\nu). \tag{3.2.30}$$

We claim that

$$\mathbf{v}_p = v^\mu (\partial_\mu)_p. \tag{3.2.31}$$

To see this, let f be an arbitrary smooth function. The coordinate representation of f is a smooth function of $x^1, \ldots, x^n$, so its ordinary multivariable differential gives

$$\mathbf{v}_p(f) = v^\mu (\partial_\mu)_p(f). \tag{3.2.32}$$

Since the two tangent vectors have the same action on every smooth function, they are the same vector. Hence

$$\mathbf{v}_p = v^\mu (\partial_\mu)_p.$$

Therefore the coordinate vectors span $T_p\mathcal{M}$.

That completes the proof. □

Thus every tangent vector can be written in the form

$$\mathbf{v}_p = v^\mu (\partial_\mu)_p. \tag{3.2.33}$$

Suppressing the explicit point label when there is no ambiguity, we conventionally write

$$\mathbf{v} = v^\mu \, \partial_\mu. \tag{3.2.34}$$

---

[17] Wald, p.16

This is the origin of the familiar "arrow" notation in a traditional coordinate basis, $\mathbf{v} = v^\mu \mathbf{e}_\mu$. In this case, we prove that the coordinate basis is therefore similar, but equivalent to the elementary basis vectors. We shall expound more on this next meeting.

$$\mathbf{e}_\mu \equiv \partial_\mu = \left. \frac{\partial}{\partial x^\mu} \right|_p$$

For example, in Cartesian coordinates on $\mathbb{R}^3$,

$$\mathbf{v} = v^x \frac{\partial}{\partial x} + v^y \frac{\partial}{\partial y} + v^z \frac{\partial}{\partial z}. \tag{3.2.35}$$

In spherical coordinates, the very same tangent vector can instead be represented as

$$\mathbf{v} = v^r \frac{\partial}{\partial r} + v^\theta \frac{\partial}{\partial \theta} + v^\phi \frac{\partial}{\partial \phi}. \tag{3.2.36}$$

The components v^μ change when we change coordinates, while the geometric vector $\mathbf{v}$ itself does not.

This is one of the central ideas of differential geometry:

> The vector is the geometric object; its components are merely its representation in a chosen basis.

3.2.8 Abstract Index Notation

In general relativity, it is useful to distinguish between a vector as an abstract geometric object and its components in a particular coordinate system.

We may therefore write

$$\mathbf{v}$$

for the vector itself, while

$$v^\mu$$

denotes its components with respect to the coordinate basis ∂_μ.

Thus,

$$\mathbf{v} = v^\mu \, \partial_\mu \tag{3.2.37}$$

The index μ is not itself a coordinate label such as x or y. It is an abstract index indicating which component of the vector is being referred to.

In a coordinate system,

$$\mu = 0, 1, \ldots, n-1,$$

and Einstein summation convention is understood whenever an index appears once upstairs and once downstairs, it is summed.

For example, in four-dimensional spacetime,

$$\mathbf{v} = v^\mu \, \partial_\mu = \sum_{\mu=0}^{3} v^\mu \, \partial_\mu = v^0 \, \partial_0 + v^1 \, \partial_1 + v^2 \, \partial_2 + v^3 \, \partial_3. \tag{3.2.38}$$

In a coordinate system

$$x^\mu = (t, x, y, z),$$

this becomes

$$\mathbf{v} = v^t \frac{\partial}{\partial t} + v^x \frac{\partial}{\partial x} + v^y \frac{\partial}{\partial y} + v^z \frac{\partial}{\partial z}. \tag{3.2.39}$$

The crucial conceptual transition is therefore

$$\boxed{\text{vector} \quad \longrightarrow \quad \text{derivative operator}} \tag{3.2.40}$$

rather than

$$\text{vector} \quad \longrightarrow \quad \text{arrow in space.}$$

The arrow remains a useful visualization, but the derivative-operator definition is what genuinely survives on an arbitrary differentiable manifold. However, for a greater understanding of the geometries involved in general relativity, where spacetime is curved, where there is no global notion of parallelism, and where vectors at different points cannot be directly compared, we need to recover the *sense* or notion of being an arrow of a vector from first principles. This recovery comes through curves: a vector is realized as the tangent to a curve, giving us an arrow with both a direction and, once a metric is introduced, a magnitude. This will be discussed at length in the coming lectures. Without this recovery, the formalism remains abstract and disconnected from the physical intuition that drives relativity.

3.3 Vectors on Manifolds

3.3.1 Review of Structures on Manifolds

Last meeting, we talked about structures on manifolds. These are the mathematical objects that will be important in general relativity. We discussed scalars, which was fairly easy, and we also started looking at vectors.

The key thing here is that a vector on a manifold is really a map: it is a derivative operator, or a derivation. At least, this is how we tried generalizing it.

So by that, we mean it's a map that can act on functions. So if you have a vector $\mathbf{v}_p$ on a point $p \in \mathcal{M}$, which belongs to $T_p\mathcal{M}$,

$$\mathbf{v}_p \in T_p\mathcal{M} \tag{3.3.1}$$

that means it can act on functions, and these actions on functions will be a real number.

$$\mathbf{v}_p : f(\mathcal{M}) \to \mathbb{R}$$

So it's kind of like a directional derivative. The intuition is that if you provide a function to $\mathbf{v}_p$, it will return a rate of change of that function along some kind of direction.

We haven't really completely filled in the details of that story. As we discussed before, we have to recover the arrow again. We ask how do we explicitly make a vector look like a directional derivative in $\mathbb{R}^n$, so that's in part what we are going to achieve today.

The other thing we discussed last time is that $T_p\mathcal{M}$, otherwise called a tangent space (we will justify why that name is appropriate), is basically the set of all vectors, or directional derivatives, at p. It's a vector space, or a linear space, but for it to become a vector space we had to define what the operation of plus, $+$, means, what scalar multiplication means, we also introduced some properties that a vector must satisfy, and similar notions of coordinate basis vectors and abstract index notation. Today, we proceed with the apt description that further investigates what vectors are.

3.3.2 Coordinate Bases

Now, being a vector space, one of the most important things one can do when one has a vector space is establish a basis on that vector space. We know already from our many experiences with vector spaces in physics. One nice basis is the so-called coordinate basis on $T_p\mathcal{M}$. The coordinate basis is one that we get from having a chart containing p.

So if we have a manifold and a point p, remember that it is going to be an open set that contains p, and then there is going to be a chart ψ that maps p to a coordinate system.

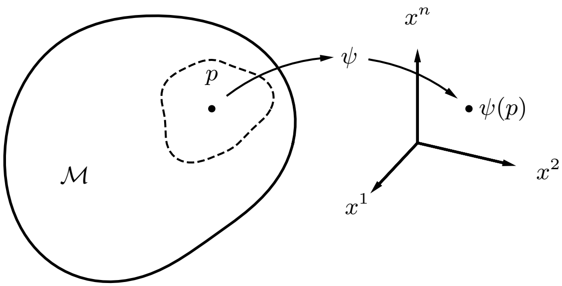


Figure 14: Mapping of p from a manifold $\mathcal{M}$ to $\mathbb{R}^n$ via ψ

This we often just write as $\{x^\alpha\}$, instead of saying ψ, which is fairly abstract; we just say $\{x^\alpha\}$ and express it in terms of the coordinate functions. $\{x^\alpha\}$ are the coordinate functions associated with ψ.

$$\psi \iff \{x^\alpha\}$$

and once we have that we have our basis, which is expressed as a directional derivative in terms of the coordinate functions.

$$\left\{ \frac{\partial}{\partial x^\alpha} \right\}$$

Often times, we will write this simply as

$$\{\partial_\alpha\}$$

When it's confusing, for example, when you have two different bases, we will write this differently. For example, if you have a different basis from a different chart $\{y^\alpha\}$, we instead explicitly define each basis vector

$$\psi_x \implies \left\{\frac{\partial}{\partial x^\alpha}\right\}, \qquad \psi_y \implies \left\{\frac{\partial}{\partial y^\alpha}\right\}$$

An example would be different representations of a vector in Cartesian coordinates in $\mathbb{R}^3$ or spherical coordinates in $\mathbb{R}^3$. As a shorthand, we can also write

$$\left\{\frac{\partial}{\partial x^\alpha}\right\} = \{\partial_\alpha\}, \qquad \left\{\frac{\partial}{\partial y^\alpha}\right\} = \{\partial'_\alpha\}$$

If there is confusion, like when there is a possibility that we're working on different charts, with two different sets of coordinate vectors, we can use this convention of having a primed operator.

We also discussed last time that since it's a basis, I can now express any vector $\mathbf{v}_p$ in terms of any coordinate basis, so we can write, in terms of the x chart,

$$\mathbf{v}_p = v_p^\alpha\,\partial_\alpha = v_p^\alpha \frac{\partial}{\partial x^\alpha}$$

Here, we said that v_p^α is really nothing but the action of $\mathbf{v}_p$ on the coordinate basis function x^α. This is what we established last time.

$$v_p^\alpha = \mathbf{v}_p(x^\alpha)$$

Question: So is ∂_α equivalent to the elementary basis vectors e_i?

No, not all the time. They are kind of similar, but they are not quite the same. The elementary basis vectors in traditional physics are equivalent to them in Cartesian $\mathbb{R}^3$, simply a different way of writing

$$\hat{\boldsymbol{\imath}}, \hat{\boldsymbol{\jmath}}, \hat{\boldsymbol{k}} \equiv \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}$$

However, in other coordinate systems, this is not necessarily the case, because when we say something like $\mathbf{e}_\alpha$, then we are already saying something about the magnitude of these vectors, since these are unit and orthogonal vectors, whereas ∂_α are not necessarily unit nor orthogonal because we do not have a notion of length here yet, and we do not have a notion of orthogonality as well. Those statements precede those notions. We can have coordinate basis vectors even without the notion of lengths, orthogonalities, or inner products. In Cartesian coordinates, these will turn out indeed to be equivalent once we introduce the so-called Euclidean metric. The Euclidean metric is what's going to provide us a sense of length and orthogonality, which is an inner product. Prior to that, without the metric, we already have coordinate basis vectors, but not the usual sense of lengths or orthogonality, which $\hat{\boldsymbol{\imath}}$, $\hat{\boldsymbol{\jmath}}$, and $\hat{\boldsymbol{k}}$ already contain.

For example, the spherical basis vectors point in the same direction as ∂_α, but they are not equivalent to the coordinate basis vectors you get from the manifold.

$$\hat{\boldsymbol{r}}, \hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\phi}} \not\equiv \frac{\partial}{\partial r}, \frac{\partial}{\partial \theta}, \frac{\partial}{\partial \phi}$$

These are still coordinate basis vectors; it's very easy—you just take the coordinate and put it in the denominator, but they have (you will see this when we introduce the meaning of length) different magnitudes. Later on, when we introduce the notion of normalization and define a unit vector, then we recover the same vectors.

3.3.3 Change of Coordinates

Now we proceed to the question that we left hanging in the last lecture.

Suppose $p \in \mathcal{M}$ belongs to two different charts, ψ_1 and ψ_2, and as such we have two different sets of coordinate functions that apply to and describe p.

If that is the case, then we also have different sets of coordinate basis vectors on $T_p\mathcal{M}$. It's the same $T_p\mathcal{M}$, but for each chart, it can be described by two different sets of basis vectors, $\{x^\alpha\}$ and $\{y^\alpha\}$. So the question becomes: how do the different sets of components of $\mathbf{v}_p$ relate from one chart to another? Note that the components v^α are not the same as the vector $\mathbf{v}$. The components can change, but the vector remains the same.

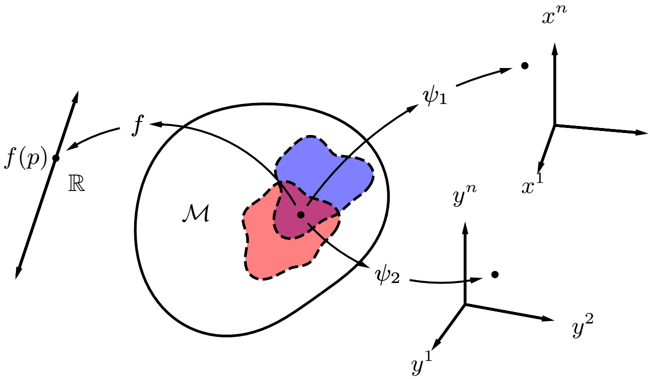


Figure 15: Representation of a point $p \in \mathcal{M}$ in coordinate basis charts $\{x^\alpha\}$, $\{y^\alpha\}$, and a scalar function f

Notice that x^α is the αth coordinate function on the chart ψ_1, since $\psi_1 : p \mapsto x^\alpha$

$$x^\alpha = \psi_1^\alpha(p)$$

Similarly for ψ_2, we have

$$y^\alpha = \psi_2^\alpha(p)$$

We asked last lecture: how do we express a coordinate transformation from y to x in terms of ψ_1 and ψ_2? We find that

$$y^\alpha(x^\beta) = (\psi_2 \circ \psi_1^{-1})^\alpha(x^\beta)$$

Now we want to understand how we define a vector present in both charts, and how its components relate to each other.

To specify a vector, as we defined it in the last lecture, is to specify a map from functions to real numbers. We first need to define the map. Every vector is a derivative operator. If you tell me how it operates on a function, then you will have specified the vector.

Suppose a vector on the y basis at point p is acting on a function. Recall how we defined this in the last lecture:

$$\left(\frac{\partial}{\partial y^\alpha}\right)_p f = \left.\frac{\partial f \circ \psi_2^{-1}}{\partial y^\alpha}\right|_{\mathbf{y}_p} \tag{3.3.2}$$

Similarly, the x coordinate vectors have exactly the same form.

$$\left(\frac{\partial}{\partial x^\alpha}\right)_p f = \left.\frac{\partial f \circ \psi_1^{-1}}{\partial x^\alpha}\right|_{\mathbf{x}_p} \tag{3.3.3}$$

Notice that the definitions are exactly the same; the only difference is that we use the corresponding chart for each coordinate system. So, how do these things relate to each other? That's what we're trying to address.

We can insert, in essence, an identity map $\psi^{-1} \circ \psi$

$$\left(\frac{\partial}{\partial x^\alpha}\right)_p f = \left.\frac{\partial f \circ \psi_2^{-1} \circ \psi_2 \circ \psi_1^{-1}}{\partial x^\alpha}\right|_{\mathbf{x}_p}$$

Then, we can just expand this out with standard multivariable calculus techniques, in particular we use the chain rule.

Question: Is $\psi \circ \psi^{-1}$ also an identity?

Yes, it is also an identity map. However, we had to do it in this particular order for this derivation because we will not get the result we desire due to the non-commutative nature of these operators.

Proceeding with the operation, we find that

$$\left(\frac{\partial}{\partial x^\alpha}\right)_p f = \left.\frac{\partial (f \circ \psi_2^{-1})}{\partial y^\beta}\right|_{\mathbf{y}_p} \left.\frac{\partial (\psi_2 \circ \psi_1^{-1})^\beta}{\partial x^\alpha}\right|_{\mathbf{x}_p} \tag{3.3.4}$$

Notice the change to $\mathbf{y}_p$. Since $f \circ \psi_2^{-1}$, as defined, is simply $f(\mathbf{y}_p)$, we evaluate from there. Now, what is $\psi_2 \circ \psi_1^{-1}$? From the last lecture, as you can verify, this is just $\mathbf{y}(\mathbf{x})$, i.e., it can be represented as y^β.

Also notice that we used another dummy index β. Since we used the chain rule, which sums over intermediate coordinates, we need to represent the intermediate coordinate index separately from the fixed α, which labels the x-coordinate we are differentiating with respect to.

Now we can represent (3.3.4) as

$$\left(\frac{\partial}{\partial x^\alpha}\right)_p f = \left.\frac{\partial (f \circ \psi_2^{-1})}{\partial y^\beta}\right|_{\mathbf{y}_p} \left.\frac{\partial y^\beta}{\partial x^\alpha}\right|_{\mathbf{x}_p}$$

But now, looking at this, we see that the first term is just (3.3.2). This is nothing but

$$\left(\frac{\partial}{\partial x^\alpha}\right)_p f = \left(\frac{\partial}{\partial y^\beta}\right) f \cdot \left.\frac{\partial y^\beta}{\partial x^\alpha}\right|_{\mathbf{x}_p}$$

We can rearrange this because this is just multiplication, and $\left.\dfrac{\partial y^\beta}{\partial x^\alpha}\right|_{\mathbf{x}_p}$ are just numbers. Then

$$\left(\frac{\partial}{\partial x^\alpha}\right)_p f = \left.\frac{\partial y^\beta}{\partial x^\alpha}\right|_{\mathbf{x}_p} \left(\frac{\partial}{\partial y^\beta}\right) f$$

Notice that this is for an arbitrary function f, and as such, in an operational sense, we can write

$$\left(\frac{\partial}{\partial x^\alpha}\right)_p = \left.\frac{\partial y^\beta}{\partial x^\alpha}\right|_{\mathbf{x}_p} \left(\frac{\partial}{\partial y^\beta}\right)_p \tag{3.3.5}$$

One of the things that you will obviously see here is that this is nothing but the expression of the chain rule in multivariable calculus. It is as if we just wrote down the chain rule. It's very easy to remember. However, note that this has a different interpretation: we are giving a new interpretation to the chain rule. Because what is this statement actually telling you? What we're doing here is expressing the left-hand side, which is a vector, as a superposition, or a sum, over other basis vectors. So we are expressing a vector on the left-hand side as a linear combination of vectors on the right-hand side.

So something like the chain rule, which is taken for granted in multivariable calculus because it is used so often, now gets a different interpretation in terms of vectors. The values on (3.3.5) are just numbers, so it's as if we are doing

$$|\psi\rangle = \psi^i \, |e_i\rangle$$

or a matrix multiplication

$$\mathbf{x} = \mathbf{M}\mathbf{y}$$

We interpret (3.3.5) as the following: The αth basis vector in the x-coordinate system is a linear combination of the basis vectors in the

y-coordinate system. Similarly, you can express

$$\left.\frac{\partial}{\partial y^{\alpha}}\right|_{p} = \left.\left(\frac{\partial x^{\beta}}{\partial y^{\alpha}}\right)\right|_{\mathbf{y}_p} \left(\frac{\partial}{\partial x^{\beta}}\right)_{p}$$

So it means that we can represent one basis in terms of the other. So how then do we relate the different components coming from these basis vectors?

What we did earlier was that, given two sets of basis vectors, we asked how to relate one in terms of the other.

By the way, the expressions $\frac{\partial y^{\beta}}{\partial x^{\alpha}}$ are Jacobians; this is often called the tangent map, which we will explain later on. This is actually the partial derivatives of the transition functions, which are defined to be smooth in smooth manifolds.

3.3.4 Vector Transformation Laws

If I have a vector $\mathbf{v}_p \in T_p\mathcal{M}$, and I have two different charts, then I get to represent this vector using two different sets of basis vectors. I can say that

$$\mathbf{v}_p = v_p^{\mu} \frac{\partial}{\partial x^{\mu}} \tag{3.3.6}$$

or we can write the same vector as

$$\mathbf{v}_p = {v'_p}^{\nu} \frac{\partial}{\partial y^{\nu}} \tag{3.3.7}$$

The important thing here is that we are talking about the same vector. Just like we can represent the vectors in different ways in different coordinate systems

$$\mathbf{v} = v_x \hat{\boldsymbol{\imath}} + v_y \hat{\boldsymbol{\jmath}} + v_z \hat{\boldsymbol{k}}$$
$$\mathbf{v} = v_r \hat{\boldsymbol{r}} + v_\theta \hat{\boldsymbol{\theta}} + v_\phi \hat{\boldsymbol{\phi}}$$

Since they are the same vector, then we can equate (3.3.6) and (3.3.7)

$${v'_p}^{\nu} \frac{\partial}{\partial y^{\nu}} = {v_p}^{\mu} \frac{\partial}{\partial x^{\mu}}$$

But we can express $\dfrac{\partial}{\partial y^\nu}$ in terms of the coordinate basis vectors in x, as we have shown

$$v'_p{}^\nu \left(\frac{\partial x^\mu}{\partial y^\nu} \frac{\partial}{\partial x^\mu} \right) = v_p{}^\mu \frac{\partial}{\partial x^\mu}$$

Though remember, the mnemonic is that of the chain rule. Functionally, this is different: it is a linear combination of the basis vectors. Proceeding, we can move the LHS to the RHS, and we find

$$\left(v'_p \frac{\partial x^\mu}{\partial y^\nu} - v_p^\mu \right) \frac{\partial}{\partial x^\mu} = \mathbf{0}$$

Now we invoke the fact that we proved last lecture: that $\dfrac{\partial}{\partial x^\mu}$ is a basis, so the vectors are linearly independent. If the sum of the coefficients and the vectors is the zero vector, then the components inside the parentheses are equal to zero. Hence

$$v'^\nu_p \frac{\partial x^\mu}{\partial y^\nu} - v_p^\mu = 0$$

As such we can write

$$v_p{}^\mu = v'^\nu_p \frac{\partial x^\mu}{\partial y^\nu} = \frac{\partial x^\mu}{\partial y^\nu} v'_p{}^\nu \tag{3.3.8}$$

Similarly, we can express (3.3.8), by inverting the argument and taking the reciprocal of the derivative as

$$v'_p{}^\nu = \frac{\partial y^\nu}{\partial x^\mu} v_p^\mu \tag{3.3.9}$$

So this is how the components of the vector $\mathbf{v}_p$ relate to one another under a change of basis; this is a component transformation.

Some of you will recognize this as the Arfken[4] standard definition of a vector. In fact, it is a contravariant vector. This is the old-fashioned definition of a contravariant vector. This is what's being taught by Arfken and a lot of other elementary or introductory physics books. We say old-fashioned because this is how it has traditionally been done before, but for us it is insufficient. For most physicists and most physics work, it is probably sufficient, but it doesn't really answer what a vector is, in a rigorous sense.

If you ask "What is a vector?", a physicist would say, "Well, if its components transform this way, it's a vector" or "It's an element of a vector space", without really saying or answering what a vector is. So now, we know what a vector is: it's a derivative operator, and as a consequence of that, its components transform in a manner described by (3.3.9).

So this is no longer the definition of a contravariant vector; it is the transformation rule arising from the fact that vectors are derivative operators. For classical physics, it is generally sufficient to say how it transforms, but the problem is in GR and, in fact, for most of physics, we are emphasizing the geometric object itself, defining what the object is rather than how its components transform under different coordinate systems.

We are trying to divorce ourselves from coordinates as much as possible. We are trying to work with the geometric objects themselves. Now we say it is a contravariant vector because there is a different kind of vector which is a covariant vector.

3.3.5 Contravariant and Covariant Vectors

A contravariant vector is as defined earlier. The appropriate transformation for such a vector is

$$v_p'^{\,\nu} = \frac{\partial y^\nu}{\partial x^\mu} v_p^\mu.$$

It is called a contravariant vector because its components transform inversely (or "contra") to the basis vectors. Recall that the basis vectors themselves transform as

$$\frac{\partial}{\partial x^\alpha} = \frac{\partial y^\beta}{\partial x^\alpha} \frac{\partial}{\partial y^\beta},$$

so the basis vectors in the new coordinate system are related to the old ones by the same Jacobian matrix. The components, however, use the inverse Jacobian:

$$v^\mu = \frac{\partial x^\mu}{\partial y^\nu} v'^\nu.$$

This "opposite" transformation behavior—basis vectors transforming one way and components transforming the inverse way—is the origin of the term "contravariant." The components are said to transform contravariantly.

In contrast, a covariant vector (or one-form) has components that transform in the same way as the basis vectors. If you have a covariant vector ω_p, the transformation rules are expressed as

$$(\omega_p)'_\nu = \frac{\partial x^\mu}{\partial y^\nu}(\omega_p)_\mu \tag{3.3.10}$$

This is the definition of a covariant vector. But later on, in a couple of lectures from now, we will see that we will no longer call these vectors. These are actually going to be one-forms. Something which transforms like (3.3.10) is called a one-form, or alternatively, a dual vector. We will explore these in much greater detail in the next meeting. For now, it suffices to recognize that contravariant vectors are what we traditionally think of as "arrows" or tangent vectors, while covariant vectors are linear functionals that act on them to produce real numbers—the natural generalization of the dot product or inner product in a coordinate-free setting.

This distinction will become crucial when we introduce metrics and begin to discuss how to raise and lower indices, compute lengths, and define angles in curved spacetime. But those are topics for future lectures

So that we aren't so confused, we have encountered these in physics before, a contravariant vector is equivalent to a *ket*,

$$|\psi\rangle$$

and a covariant vector is equivalent to a *bra*

$$\langle\phi|$$

We do not say that bras and kets are the same. Of course, they are isomorphic, but they are quite different. They serve different purposes. Covariant and contravariant vectors are usually defined in terms of how they transform in physics, but we will define these other vectors later on, when we discuss one-forms.

3.3.6 Non-Coordinate Bases

Now it's important to emphasize that $T_p\mathcal{M}$ is a vector space. You don't always need coordinate basis vectors for vector spaces. In fact, later on we will find that it is quite practical and useful to adopt non-coordinate basis vectors.

So the vector basis need not be coordinate basis vectors; in other words, they don't need to come from a coordinate system or a chart because it's a space, you can put in a plethora, an infinite number of basis vectors on it.

For example, you have a set of basis vectors; you will need n of them if you have an n-dimensional manifold, and they will have some components with respect to some basis, and these bases need not come from a coordinate system; they can be linear combinations of coordinate basis vectors.

$$e_{(\beta)} = e^{\alpha}_{(\beta)} \frac{\partial}{\partial x^{\alpha}}$$

The only constraint is that it must have a nonzero determinant. You don't get a basis unless the matrix formed by the basis vectors has a nonzero determinant. It must be nonsingular; otherwise, you do not have a basis.

Later on, when we go to Lie derivatives, we will investigate under what conditions $e_{(\beta)}$ will actually have a coordinate representation. The short answer is that the Lie derivative should be zero.

$$\mathscr{L}_{(\beta)} e_{(\alpha)} = 0$$

If this is true for any α and β for this new set of basis vectors that we've introduced, then there is a set of coordinates for which these are coordinate basis vectors. I can find a coordinate system to write, for example:

$$e_{\beta} = \frac{\partial}{\partial z^{\beta}}$$

If the Lie derivative condition is satisfied, but we'll get to that much later on.

Recovering the Geometric Arrow So now, we want to be able to recover the notion of an arrow in $\mathbb{R}^n$. So far, we've had a lot of properties that we have sort of found and recovered in what we understand to be traditional vectors in physics. But to really get a sense of an arrow, we need to talk about curves and tangent vectors.

3.4 Curves and Tangent Vectors

3.4.1 Curves

We feel like we have a notion or sense of what a curve is, since we encounter it in physics a lot. However, for our purposes, we need to precisely define what a curve is.

Definition 3.7 : Curves

A curve C in a manifold $\mathcal{M}$ is a map from an open interval $I \subset \mathbb{R}$ onto $\mathcal{M}$

$$C : I \to \mathcal{M} \tag{3.4.1}$$

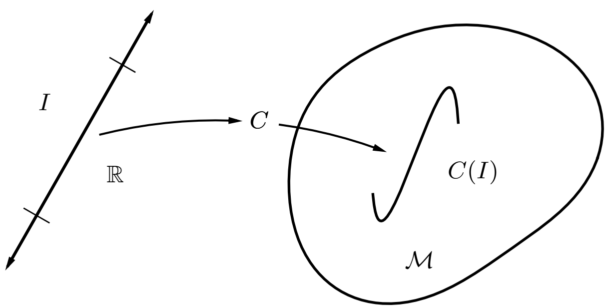


Figure 16: A curve mapping a real line interval to a manifold

$C(I)$ is a function, which is different from how we usually imagine a curve, which is some line with a curvature on R^n. $C(I)$ is not actually the curve itself, but the image of the curve. So what we tend to think of as the curve is really the image of the curve C, because C itself is the map; C is the function. It's the instruction manual that tells us how to relate a real number to a point in $\mathcal{M}$, i.e., $C : \mathbb{R} \mapsto p$. $C(I)$ is a set of points in $\mathcal{M}$. And $I \subset \mathbb{R}$ is called a parameterization. I need not be a proper subset; it can be the entire real line. It is how we parameterize a curve. Say, for example, you want to think of the parameter I as time, t, which is a very typical parameter in physics; the parameter need not be uniform. People can have different parameterizations, different clocks running at different rates, for instance.

Imagine a different curve, say $C' : I' \to \mathcal{M}$, with different parameters I', such that

$$C(I) = C'(I')$$

then we call these curves to have the same image, but different parameterizations. So we often say that C and C' are reparameterizations of each other. They have the same image, the same set of points, but the parameter I that maps it is different, i.e., $C'(t') \neq C(t)$.

If they are reparameterizations of each other, then it is not so strange that there will exist an onto map, a surjective map $\alpha : I \to I'$, such that

$$\alpha(t) = t'$$

and

$$C = C' \circ \alpha$$

An important thing to emphasize here is that a curve on its own does not need coordinates. It's a map from the subset of the real line to the manifold. However, in physics, it's very common to express curves in terms of coordinates, which we can do on manifolds by mapping the points $C(I)$ to $\mathbb{R}^n$. Pictorially, we can represent this as

As such, we can recover the normal parameterization of a real number as a coordinate in $\mathbb{R}^n$.

$$\psi \circ C : I \to \mathbb{R}^n$$

Recall that $I \subset \mathbb{R}$, therefore $\psi \circ C : \mathbb{R} \to \mathbb{R}^n$. In other words

$$\psi \circ C : t \mapsto x^\alpha(t)$$

So for example, $[x(t), y(t), z(t)]$, which is how we normally describe curves in physics. You can provide a coordinate representation of curves, but the curve itself actually did not require the coordinates. So what we do is coordinatize a curve.

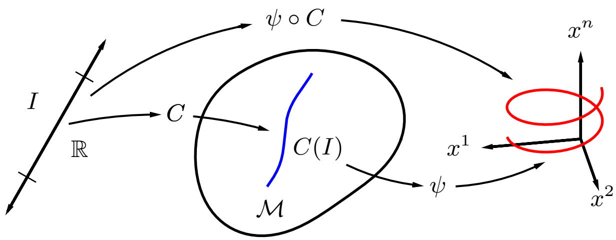


Figure 17: $\psi : C(I) \to \mathbb{R}^3$

3.4.2 Coordinate Curves

There are going to be these special curves which we need to define. Given some chart, which we represent with $\{x^\alpha\}$, consider the following subsets of open sets in $\mathcal{M}$ for which the coordinates are valid, with chart ψ, and atlas $\{\mathcal{O}, \psi\}$. Consider the following subset of all points in $\mathcal{O}$ such that

$$\left\{p \in \mathcal{O} : x^2(p) = \text{constant},\ x^3(p) = \text{constant},\ \ldots x^n(p) = \text{constant}\ \right\}$$

What do you imagine these sets of points to be? The image of this map is a straight line at $x^1(p)$, because we are keeping all the x^α's constant, except for x^1. However, the set that we're looking at, since it's a set of all p's in $\mathcal{O}$, is the pre-image of the points that will get mapped into a line. So this set is asking you to imagine what points in $\mathcal{O}$ will get mapped into a straight line by ψ.

So with this set, consider the inverse map ψ^{-1}; it defines a set of points that will get mapped to a straight line. Since we know it is continuous, then that necessarily means that the pre-image is a continuum of arbitrary points, i.e., the preimage is a curve on the manifold, because it's mapping an open interval of $\mathbb{R}$ onto a set of points in $\mathcal{M}$. This is a special type of curve; it's called the x^1 coordinate line or x^1 coordinate curve. Notice here that the parameter is x^1. This is an important notion.

The x^1 coordinate curve is the image of a curve on the manifold whose parameter is the x^1 coordinate.

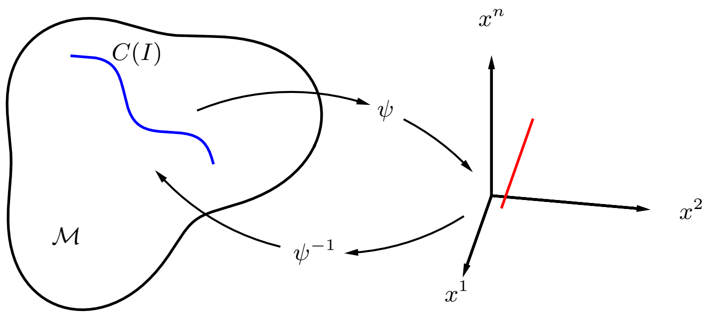


Figure 18: Curve on $\mathcal{M}$ mapped to $\mathbb{R}^n$, resulting on a curve with only x^1 varied, and the rest of x^α constant

We can then imagine, instead, doing the same for a line where x^2 is a parameter, then x^3, and so on, until x^n. What you then form then is a map of the correspondences of each axes or coordinate basis in the chart to a curve on the manifold itself, a set of curves on the manifold that correspond to each coordinate axes of the chart. You are therefore, placing a reference grid on how points on the manifold will map to $\mathbb{R}^n$ when ψ is applied. Each point p on the chart lies at the intersection of exactly n coordinate curves, one for each coordinate direction. The coordinate curves depend entirely on the choice of chart. A different chart will generally produce a different set of curves on the manifold

A second property is that the tangent vector, which is a vector that we will define later for curves, of the coordinate curves form a coordinate basis. The tangent vector to the x^α coordinate curve at p is precisely the coordinate basis vector ∂_α.

If you imagine a manifold as a curved surface embedded in higher-dimensional space, then coordinate curves are the grid lines drawn on the surface. The tangent vectors to these curves are the arrows that point along the grid lines. Any tangent vector can then be thought of as a linear combination of these grid-line arrows.

3.4.3 Tangent Vectors at Curves

You have an intuitive sense of what a tangent vector is: it's the arrow that points in the direction instantaneously of the curve. In a sense, a tangent vector is our traditional notion of what a tangent vector is. But because it's a vector, we now know that it's supposed to be a directional derivative. We have to appropriately define, then, a tangent vector as a derivative operator.

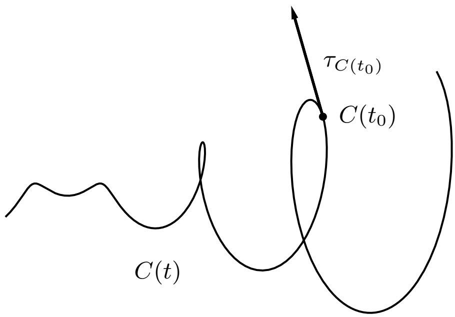


Figure 19: Tangent vector τ on a curve $C(t)$

It's always associated with a curve; i.e., if there's a curve, there must be a tangent vector. We define it then as such.

Definition 3.8 : Tangent Vector at Curves

If $C(t)$ is a curve, then the tangent vector τ at $C(t_0)$ is the derivative operator such that $\tau_{C(t)}$ acting on an arbitrary function f, is defined to be

$$\tau_{C(t_0)}(f) = \left.\frac{d(f \circ C)}{dt}\right|_{t_0} \tag{3.4.2}$$

Now we ask, what is the nature of the object that results from this definition? What are its properties? What is the map $f \circ C$? Remember that f is a map from the manifold to real numbers, and C is a map from real numbers to the manifold. So $f \circ C$ is a map from real numbers to real numbers.

$$f \circ C : \mathbb{R} \to \mathbb{R}$$

So it's simply a normal function, a perfectly sensible real function, which is a derivative with respect to a parameter.

Notice that the definition of a tangent vector requires a curve. If you have the same curve C, but evaluate it at another parameter t_1, then it makes a different tangent vector acting on the same function, but evaluated at a different point.

$$\tau_{C(t_1)}(f) = \left.\frac{d(f \circ C)}{dt}\right|_{t_1}$$

This is a different tangent vector. We can show that this is in fact a vector as we previously defined it, i.e., it must satisfy linearity and the Leibniz property, but by the nature of the function, since it's a derivative operator, it obviously satisfies both. This definition satisfies the properties of being a vector, or being a derivation. We will recover the notion of an arrow from this; this is where the traditional arrow definition of a vector comes from.

Here, we only define it. We shall discuss it in detail in the next meetings.

Chapter 4

Einstein Field Equations

Chapter 5

Matter Models

Chapter 6

Weak-field Solutions

Chapter 7

Spherical Symmetry

Chapter 8

Basic Cosmology

Additional References

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Author's Note

A grandiose essay

My journey in learning relativity is quite curious. When I was a child, I remember wondering why things go down the way they do. My folks and those who are older would simply answer: That's just the way things are, it's gravity. *Basta.* Later on, when I was a little older, I picked up this old book called *Science in Daily Life* where I learned much about science and chemistry and all those good shit. That book was one of my prized possessions, I still have it today, in fact. When I was a little older, right around fourth grade, my neighbor was throwing out books she no longer used, and part of that mixture were books on chemistry and books on Physics for fourth year high school.

It was there where I learned about Newton and his apple. It was there where I learned about Einstein and binding energies and logic gates, of nuclear physics and atom bombs; where I learned that gravity is a force that drew upon it other masses by the product of their quantities the inverse of the square of their distances. That made me interested in orbital mechanics, astronomy, the Apollo program. That book was one of the reasons why I am where I am today, the same way Vsauce and Veritasium inspired me when I was watching their videos when I was young. I say all this, because we all have an origin story of some sort. Some story we call ours, that tells our own humble beginnings. This was a story of how I fell in love with Physics, and why I chose it to be my career once I reached adulthood. That's why I decided to be a Physicist.

Newton had origin stories. Einstein had an origin story. We must treasure where we came from, and what it took for us to get there, is all I'm trying to say. Anyways, I first learned about general relativity all those years ago, but thought it was very difficult and a hassle to understand. Understanding special relativity and its implications were much easier, but boy, GR is a whole 'nother level. But fascinating! Differential geometry, manifolds, paradoxes, black holes, all so fascinating and interesting for me, part of the reason why I fell in love with the damn philosophy of reason. But also, it was scary, all those numbers and derivatives and the like, all so intimidating, that I thought back then I would never even learn it. And so, I find it funny, now, as an adult, that I took it as my undergraduate specialty of choice, joining a laboratory that specialized in gravity, taking my thesis on gravity. The amount of difficult mathematics one had to master to reach this point is insane, but, much like a mountain, the difficulty is part of the journey, and to reach the top is sweeter than the pleasures we get to enjoy in our daily life.

And so, here I am today, taking a graduate course in my undergraduate degree, about the subject that fascinated me as a child, but I found too difficult to climb and learn. Now, I know enough to say that what I know is a drop in an endless ocean, and relativity, the jewel of modern physics, in both beauty and elegance, is the pearl at the bottom of it. It makes me glad that it is what I chose to dedicate my academic life to, and I find within me the desire to add to the sum of human knowledge, to participate in the accumulated species-wide project of humanity towards understanding nature, to leave a legacy behind and be remembered in such a way. It is only recently that I learned to appreciate physics for what it is, a beautiful description of reality. In my early college years, the stress of it all consumed me, so much so that I haven't really taken a break and looked at the landscape, and now I am about to graduate, I look back think of all those things I learned, and it impresses me that I could learn so much.

Anyways, these lecture notes are precisely that, transcriptions of lectures on the course of General Relativity, by one of the leading physicists in the country, Dr. Vega, whom I personally idolized since I learned about him when I was in highschool, being a homegrown gravitational physicist and all that, and now I'm studying under him as my adviser, ain't that something. The content is precisely what he talks about in the lecture. Although not exactly verbatim, I tried my best to record all the words being said and tried to capture the topic being discussed at the moment, doing my best to convey all the ideas and meaning of the lecture. So that, in the future, I may look upon it when I forget something or simply as I please, and others, peers, students, friends, which I may give these lecture notes to, may understand it in a manner as I understand it today. I wish the future may be kind to us, and when we look back, to our stories and origins, may we smile and reminisce in mild happiness, and be proud of what we have become.

End.

PHYSICA VINCIT OMNIA

Fideli meo computatro Lenovo 81AH, comite laboris, confectum
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Augusto MMXXVI

Notæ Lectionum de Relativitate Generali

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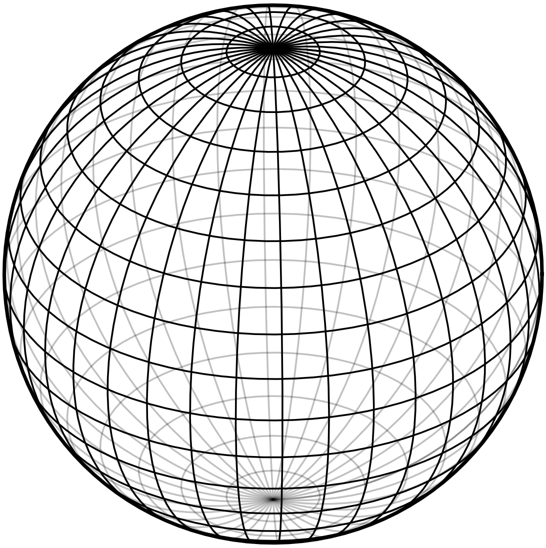

Sic Mundus Creatus Est

Ad Astra per Aspera

Hic sum, hic maneo

Mors vincit Omnia